An Empirical Model of Dynamic Two-Sided Matching: Application to the Teacher Labor Market*

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Abstract

This paper develops an empirical model of dynamic two-sided matching with non-transferable utility where matches are stable in each period and agents have rational expectations about their future rematching opportunities. I consider a large market approximation and prove existence and uniqueness of the limiting distribution of matched characteristics. I show that the surplus function is identified from data on realized matches and provide sufficient conditions for separate identification of individual preferences. I develop a tractable estimation procedure and apply these tools to study the dynamics of the teacher labor market in Peru. Teacher mobility exacerbates urban-rural disparities in student achievement as high value-added teachers increasingly sort in urban areas throughout their career. Dynamic contracts mandating a minimum spell length in rural schools to reduce this outflow would require a substantial increase in compensation to negate adverse selection.

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1 Introduction

Empirical models of two-sided matching with non-transferable utility (NTU) have played a pivotal role in evaluating various mechanism design choices and policy interventions in centralized clearinghouses allocating jobs, college seats or public housing (Agarwal, 2017; Roth, 2018; Agarwal and Budish, 2021). Many of those studies rely on the assumption that agents participating in these markets interact in a static environment, either by considering that they are myopic or that matches are irreversible. Yet, matching markets often provide participants the opportunity to delay or reconsider their matching decisions. As both preferences and opportunities evolve over time, students switch schools or majors, workers change jobs, households move. Understanding the dynamic trade-offs involved in these decisions is essential to predict agents' behavioral responses to any given policy intervention.

This paper develops an empirical model of dynamic two-sided matching with NTU where forward-looking agents repeatedly meet and (re-)match. I assume that matches are stable in each period with respect to agents' lifetime utility and that agents have rational expectations about their future rematching opportunities. I consider a large market approximation and show existence and uniqueness of the limiting distribution of matched characteristics. I leverage this characterization to show that the surplus function is identified from data on realized matches and discuss how additional data can be leveraged to separately identify individual payoff functions. I then provide a tractable estimation procedure and apply these tools to study the dynamics of the teacher labor market in Peru. Teacher mobility exacerbates urban-rural inequality in student achievement as high value-added teachers starting their careers in rural areas are more likely to exit the profession or move to urban schools than low value-added teachers. I show that dynamic contracts mandating a minimum spell length in rural schools to reduce this outflow would require a substantial increase in compensation to negate adverse selection.

To conduct this analysis, I consider a one-to-one matching model with many agents. I refer to one side as firms and the other side as workers. Agents are forward-looking and form preferences over an arbitrarily large set of worker and firm attributes as well as the continuation value they would get from any given match. Once a match is formed, firm and worker attributes evolve endogenously according to their matching decisions. Agents are then able to rematch in the next period. The model can accommodate the existence

of moving or switching costs as observed attributes can contain characteristics of agents' previous match. I impose two main assumptions on preferences: (i) the systematic and unobserved part of the flow payoff functions are additively separable and (ii) the unobserved taste shocks are *iid* with a type-I upper tail. I assume that matches are stable in each period with respect to agents' expected lifetime utility. Agents do not have perfect foresight about their future preferences and opportunities but instead form rational expectations about their future expected match value.

Stability maps workers' and firms' preferences into equilibrium matches summarized by the joint distribution of the characteristics of matched worker-firm pairs in each period. Under stability, each worker is matched to their preferred firm among the set of firms that would be willing to hire them. Similarly, each firm is matched to its preferred worker among the set of workers that would be willing to work with them. Realized matches are thus the outcome of single-agent dynamic discrete choice problems, with the caveat that agents' current and future choice sets, called *opportunity sets*, are unobserved equilibrium objects. Additionally, several stable equilibria could exist in each period such that rational expectations about future matching opportunities are not unique if agents' beliefs about the equilibrium selection mechanism are not specified.

To circumvent these issues, I consider a large market approximation where the number of workers and firms approaches infinity. In the limit, the information contained in opportunity sets necessary to characterize conditional choice probabilities and expected future match payoffs can be summarized by sufficient statistics called *inclusive values*. I show that stability and rational expectations imply that inclusive values are approximated by the solution of a fixed point problem. Dynamics magnify the dimensionality and complexity of this fixed point problem relative to the static case as workers' current inclusive values depend both on their current *and* future opportunities, which themselves depend on firms' current and future preferences and inclusive values. I find that there exists a unique solution to this fixed point problem such that, in the limit, conditional choice probabilities and rational expectations about future match payoffs are unique irrespective of which stable match is selected. This implies existence and uniqueness of the limiting distribution of matched characteristics, which can be expressed as a function of workers' and firms' payoff functions and inclusive values.

I leverage this characterization to show that the mapping between workers' and firms' preferences and the joint distribution of matched characteristics is invertible. I find that

the systematic part of the joint surplus function is nonparametrically identified from data on realized matches. However, individual preferences cannot be separately identified from the surplus without further restrictions. I show that additional information on the opportunity sets of agents on one side of the market is sufficient to recover separate identification of individual preferences. Finally, I propose a maximum likelihood estimator that can be tractably used in a parametric version of this framework. I then discuss an extension of this estimation strategy allowing for time-invariant unobserved preference heterogeneity.

Finally, I apply these tools to study the assignment of public teachers in Peru. I combine several administrative datasets to create a comprehensive record of all applicants and job postings participating in the nationwide centralized assignment mechanism over the 2015-2019 period. I first document that high value-added teachers increasingly sort in urban schools as they advance in their careers. Specifically, high value-added teachers starting their careers in rural areas are more likely to exit the profession or move to urban areas compared to low value-added teachers. On the other hand, high value-added teachers starting their careers in urban areas are more likely to stay in urban areas. As a result, the observed urban-rural gap in teacher value-added implied by teachers' initial assignment increases by 23% after two years.

I leverage additional information on schools' opportunity sets to separately identify and estimate teachers' and schools' preferences. Teachers face dynamic trade-offs induced by switching costs from urban to rural areas and differential returns to experience when working in urban vs. rural schools. The estimated preference parameters indicate that (i) teachers have a large distaste for the disamenities provided in rural schools and (ii) schools highly value observed measures of teacher quality, such as experience and teacher evaluation scores. Estimates of transition processes further show that high value-added teachers have larger returns to experience, measured by evaluation score gains over time, than low value-added teachers. This results in the existence of a spatial job ladder, whereby high value-added teachers transition from the bottom rung (rural schools) to the top rung (urban schools) at a faster rate than low value-added teachers. Overall, this implies that rural schools fail to retain skilled teachers and sustain disproportionately low levels of teaching experience and effectiveness.

To mitigate this effect, I investigate the effectiveness of dynamic contracts aimed at reducing teacher attrition in rural areas. I simulate the equilibrium response to a policy

that would mandate a minimum contract length for teachers accepting a position in a rural school. In a static model, agents would be myopic and would not react to the introduction of this policy. Accounting for dynamic incentives, I find that this policy creates adverse selection as high value-added teachers purposefully avoid accepting a job in a rural area since they anticipate that they will have to wait longer to rematch. I quantify the compensation that these teachers would need to receive in order to stay in their original matches. I find that this would cost the equivalent of a monthly wage increase of 19% per teacher for a moderate decrease in the urban-rural gap in teacher value-added of 0.02σ. This suggests that incentivizing retention might be too costly when the opportunities to rematch are high, which is especially true for early career high value-added teachers.

This paper contributes to several strands of the literature. Existing work relying on stability to characterize equilibrium matches in dynamic two-sided matching models have either relied on restrictions on the set of possible stable equilibria or on agents' beliefs about the equilibrium selection mechanism (Baccara and Yariv, 2021; Doval, 2022). This paper builds on earlier work studying large matching markets with NTU in static environments (Dagsvik, 2000; Menzel, 2015) to show existence and uniqueness of the limiting joint distribution of match characteristics while allowing for multiplicity in realized stable matches in each period and in agents' beliefs about future realizations of stable matches.

A large literature studying the empirical content of static two-sided matching models with NTU has established how preferences of participating agents can be identified from reported preferences (Fack et al., 2019; Agarwal and Somaini, 2020) or realized matches (Menzel, 2015; Diamond and Agarwal, 2017; He et al., 2024; Agarwal and Somaini, 2022; Ederer, 2025). Related work has further studied how one can recover preferences of agents participating in centralized priority-based assignment mechanisms while accounting for the dynamic trade-offs induced by repeated participation or by the sequential nature of the assignment mechanism (Agarwal et al., 2021; Waldinger, 2021; Huitfeldt et al., 2024; Larroucau and Rios, 2020; De Groote et al., 2025). This paper fills a gap in this literature by studying the empirical content of two-sided matching models with NTU that feature unobserved preference heterogeneity and dynamic incentives on *both* sides of the market. I show that the systematic part of the match surplus function can be nonparametrically identified from data on realized matches and discuss how one can leverage additional data to separately identify individual preferences. I then provide a tractable maximum likelihood estimator for

a parametric version of this framework.

Finally, I contribute to a literature studying teacher sorting and its implication for student outcomes through the lens of static two-sided matching models (Boyd et al., 2013; Biasi et al., 2021; Bates et al., 2025; Bobba et al., 2021). Accounting for the dynamic trade-offs involved in teachers' matching decisions is necessary to evaluate policies affecting their continuation value. I illustrate this by quantifying the equilibrium effects of mandating a minimum contract length in rural schools to reduce attrition and inequality. A static model would assume that agents are myopic and would thus not react to the introduction of this policy. Instead, I find that a substantial increase in teacher compensation would be required to negate the adverse selection caused by the implementation of this policy. Overall, studying the teacher labor market through the lens of dynamic two-sided matching models allows us to assess the relative effectiveness of a wider range of policies besides flat wage increases (Bobba et al., 2021) in improving equity and efficiency of teacher labor markets.

2 Model

Throughout this section, I refer to one side of the market as workers and the other side as firms. Workers are indexed by $i \in \mathcal{I} = \{1, ..., n_w\}$ and firms are indexed by $j \in \mathcal{J} = \{1, ..., n_m\}$. I denote firms' outside option as $i = 0$ and workers' outside option as $j = 0$. Time is discrete and indexed by $t \in \mathcal{T} = 1, ..., T$. For notational simplicity, I assume that the set of participating workers and firms is fixed over time.[1] Firms and workers repeatedly meet over time and form one-to-one matches summarized by the function μ_w, which maps $\mathcal{I} \times \mathcal{T}$ to $\mathcal{J}$ and the function μ_m, which maps $\mathcal{J} \times \mathcal{T}$ to $\mathcal{I}$. For example, $\mu_w(i, 2) = j$ means that worker i is matched with firm j in period 2. I refer to a matching as μ which collects μ_m and μ_w and summarizes the matching outcome of each agent in each period.

2.1 Dynamic Environment

State Variables and Transitions: Each worker $i = 1, ..., n_w$ and firm $j = 1, ..., n_m$ is characterized in each period t by their observed attributes, represented by two vectors of random variables x_{it} and z_{jt}. I define the probability distribution functions of x_{i1} and z_{j1} as $w_1(x)$

[1] I discuss how the asymptotic results extend to a setting where agents enter and exit exogenously in Section 3.1.

and $m_1(z)$ with support $\mathcal{X}$ and $\mathcal{Z}$, respectively. Workers' and firms' matching decisions influence how their observed attributes evolve over time through a Markov process of order one. I assume that x_{it+1} is drawn conditional on x_{it} and the attributes of worker i's matched firm in period t denoted as $z_{\mu_w(i,t)t}$. Similarly, z_{jt+1} is drawn conditional on z_{jt} and $x_{\mu_m(j,t)t}$. I denote these transition distribution functions as $w(x_{it+1}|x_{it}, z_{\mu_w(i,t)t})$ and $m(z_{jt+1}|x_{\mu_m(j,t)t}, z_{jt})$, respectively. I denote $w(x_{it+1}|x_{it}, z_{0t})$ and $m(z_{jt+1}|x_{0t}, z_{jt})$ the transition probability distribution functions for workers and firms choosing to stay unmatched in period t. Note that x_{it} and z_{jt} can contain characteristics of worker i's and firm j's previous match such that the model can accommodate for the presence of moving or switching costs.

Matching Frequency Distributions: Given a realized match μ, observed sorting patterns in each period are summarized by the matching frequency distribution functions F_{nt} measuring the expected cumulative number of workers with observable characteristics x matched with firms with observable characteristics z in period t as

$$F_{nt}(x, z|\mu) = \frac{1}{n} \sum_{i=0}^{n_w} \sum_{j=0}^{n_m} \mathbb{P}(x_{it} \leq x, z_{jt} \leq z, \mu_w(i,t) = j \text{ or } \mu_m(j,t) = i). \tag{1}$$

The function F_{nt} is normalized by n which controls the asymptotic sequence and relates to the number of workers and firms n_w and n_m in a way that will be specified in Section 2.3. As a convention, I encode the characteristics z of the outside firm and the characteristics x of the outside worker as " $*$ ". I normalize their values to $+\infty$ such that $F_{nt}(x, *|\mu)$ and $F_{nt}(*, z|\mu)$ correspond to the frequency distribution function of the characteristics of unmatched workers and firms, respectively. I then denote F_t the limit of the distribution functions F_{nt} as the size of the market n grows to infinity. I also define the joint density function of matched characteristics f_t as the Radon-Nikodym derivative of the limiting measures F_t.

Aggregate State Transitions: Individual matching decisions in period t and state transitions aggregate to form the distributions of worker and firm attributes in period $t+1$ denoted as $w_{t+1}(x|\mu)$ and $m_{t+1}(z|\mu)$, respectively:

$$w_{t+1}(x|\mu) = \int_{\mathcal{X}} \int_{\mathcal{Z}} w(x|s,h) dF_{nt}(s,h|\mu) + \int_{\mathcal{X}} w(x|s,z_0) dF_{nt}(s,*|\mu),$$

$$m_{t+1}(z|\mu) = \int_{\mathcal{X}} \int_{\mathcal{Z}} m(z|s,h) dF_{nt}(s,h|\mu) + \int_{\mathcal{Z}} m(z|x_0,h) dF_{nt}(*,h|\mu).$$

I do not impose stationarity of the aggregate state distributions.

2.2 Preferences

Agents are forward-looking and anticipate how their current matching decision might affect their future utility. In particular, they internalize that their state variables might evolve differently according to whom they match with today, which might influence their future preferences or matching opportunities. For any given $t = 1, ..., T$, I define the lifetime utility that worker i expects from being matched with firm j as

$$U_{ijt} = U_t(x_{it}, z_{jt}) + \sigma\eta_{ijt} + \beta\bar{U}_{ijt},$$

and the lifetime utility that firm j expects from being matched with worker i as

$$V_{ijt} = V_t(x_{it}, z_{jt}) + \sigma\epsilon_{ijt} + \delta\bar{V}_{ijt}.$$

This expected lifetime utility is the sum of a *flow utility* and a *continuation value*. The flow utility includes a systematic part $U_t(x_{it}, z_{jt})$ and $V_t(x_{it}, z_{jt})$, as well as additively separable preference shocks η_{ijt} and ϵ_{ijt}. The functions U_t and V_t capture non-parametrically both vertical and horizontal firm and worker differentiation induced by the observed attributes x_{it} and z_{jt}. For example, this could capture that workers value wages and amenities and differ in how they trade-off these attributes according to their characteristics x_{it}. This could also capture that firms value workers' human capital or skills, summarized by x_{it}, differently according to their technology, summarized by z_{jt}, to potentially capture production complementarities. The continuation value is the lifetime utility agents expect to get from their match in period $t+1$ conditional on matching with j in period t, denoted as $\bar{U}_{ijt}$ and $\bar{V}_{ijt}$, discounted by a per-period factor of β for workers and δ for firms. I impose the following assumptions on the flow utility:

Assumption 1 *(i) U_t and V_t are uniformly bounded in absolute value and $p \geq 1$ times differentiable with uniformly bounded partial derivatives in $\mathcal{X} \times \mathcal{Z}$ for all t.*
(ii) ϵ_{ijt} and η_{ijt} are drawn independently from x_{it} and z_{jt} from a distribution with absolutely

continuous c.d.f. $G(s)$ *and density* $g(s)$*. The upper tail of the distribution* $G(s)$ *is of type I with auxiliary function* $a(s) := \frac{1-G(s)}{g(s)}$*.*

Assumption 1.(i) is a standard smoothness and boundedness condition on the functions U_t and V_t. Assumption 1.(ii) imposes that the distribution of the unobserved shocks G belongs to the domain of attraction of the Gumbel distribution (Resnick, 1987).[2] Specifying the distribution of unobservables has been shown to be necessary for identification in single-agent dynamic discrete choice models (Magnac and Thesmar, 2002). Assumption 1.(ii) does not directly specify the distributions of ϵ and η but instead imposes a shape restriction on their upper-tail. This class of distributions encompasses most of the parametric distributions traditionally used in discrete choice models. For the Gamma distribution or the Gumbel distribution, Assumption 1.(ii) holds for $a(s) = 1$. For the standard normal distribution, it holds for $a(s) = \frac{1}{s}$.

Note that Assumption 1.(ii) also imposes that unobserved preference shocks are independent over time.[3] Prior work on dynamic discrete choice models and dynamic games has highlighted the importance of relaxing this assumption by allowing for additional forms of unobserved preference heterogeneity.[4] Although the asymptotic results discussed in Section 3 hold irrespective of whether x and z are observed or unobserved, I discuss in Section 4.3 how one can adapt existing methods to estimate this model when x and z contain time-invariant unobserved attributes.

I do not yet impose restrictions on agents' continuation values $\bar{U}_{ijt}$ and $\bar{V}_{ijt}$ and defer this discussion to the exposition of the equilibrium concept in Section 2.4. In anticipation, I define workers' *rational expectations* at time t about their future matching lifetime utility, conditional on a set of beliefs about the future matching frequency distributions, as follows:

$$\bar{U}_{ijt}(\mu) := \mathbb{E}_{(\mu,t)}\left[\sum_{\tau=t+1}^{T} \beta^{\tau-t-1}(U_\tau(x_{i\tau}, z_{\mu_w(i,\tau)\tau}) + \sigma\eta_{i\mu_w(i,\tau)\tau})|x_{it}, z_{jt}\right], \tag{2}$$

[2]This implies that there exists an auxiliary function $a(s)$ such that

$$\lim_{s\to+\infty} \frac{1-G(s+a(s)v)}{1-G(s)} = \exp(-v).$$

Within this class of distributions, we further impose that $a(s) := \frac{1-G(s)}{g(s)}$.

[3]This is akin to the conditional independence assumption of Rust (1987).

[4]See Aguirregabiria and Mira (2010) and Aguirregabiria et al. (2021) for exhaustive literature reviews of this topic.

where the expectation is taken over the future values of worker i's attributes $x_{it+1}, ..., x_{iT}$, worker i's matched firm attributes $z_{\mu_w(i,t+1)t+1}, ..., z_{\mu_w(i,T)T}$, and worker i's unobserved preference for their matched firm $\eta_{i\mu_w(i,t+1)t+1}, ..., \eta_{i\mu_w(i,T)T}$. The expectation operator $\mathbb{E}_{(\mu,t)}[\cdot]$ is indexed by μ to indicate that it is derived from a set of beliefs over the future matching frequency distributions $F_{nt+1}(x, z|\mu), ..., F_{nT}(x, z|\mu)$. Similarly, I define for firms:

$$\bar{V}_{ijt}(\mu) := \mathbb{E}_{(\mu,t)}\left[\sum_{\tau=t+1}^{T} \delta^{\tau-t-1}(V_\tau(x_{\mu_m(j,\tau)\tau}, z_{j\tau}) + \sigma\epsilon_{\mu_m(j,\tau)j\tau})|x_{it}, z_{jt}\right], \tag{3}$$

where the expectation is taken over the future values of firm j's attributes $z_{jt+1}, ..., z_{jT}$, firm j's matched worker attributes $x_{\mu_m(j,t+1)t+1}, ..., x_{\mu_m(j,T)T}$, and firm j's unobserved preference for its matched worker $\epsilon_{\mu_m(j,t+1)jt+1}, ..., \epsilon_{\mu_m(j,T)jT}$. Throughout the rest of the paper, I refer to an arbitrary vector of continuation values as $\bar{U} := (\bar{U}_{ijt})_{(i,j,t)\in\mathcal{I}\times\mathcal{J}\times\mathcal{T}}$ and $\bar{V} := (\bar{V}_{ijt})_{(i,j,t)\in\mathcal{I}\times\mathcal{J}\times\mathcal{T}}$ and a vector of continuation values under rational expectations as $\bar{U}(\mu) := (\bar{U}_{ijt}(\mu))_{(i,j,t)\in\mathcal{I}\times\mathcal{J}\times\mathcal{T}}$ and $\bar{V}(\mu) := (\bar{V}_{ijt}(\mu))_{(i,j,t)\in\mathcal{I}\times\mathcal{J}\times\mathcal{T}}$.

2.3 Asymptotic Sequence

Asymptotic arguments in two-sided matching markets require specifying an asymptotic sequence that ensures that the limiting economy retains features of the finite economy. In particular, asymptotic arguments that make the size of both sides of the market grow to infinity (Dagsvik, 2000; Menzel, 2015; Ederer, 2025) require additional normalizations to ensure that the share of unmatched individuals does not become degenerate in the limit. I normalize the flow utility of the outside option as follows

$$U_{i0t} = \sigma \max_{k=1,...,J} \eta_{i0,k} + \beta\bar{U}_{i0t} \quad \text{and} \quad V_{0jt} = \sigma \max_{k=1,...,J} \epsilon_{0j,k} + \delta\bar{V}_{0jt},$$

and impose the following assumptions:

Assumption 2 *The asymptotic sequence is controlled by $n = 1, 2, ...$ and we define:*
(i) $n_w = [\exp(\gamma_w)n]$, $n_m = [\exp(\gamma_m)n]$
(ii) $J = [n^{1/2}]$
(iii) $\sigma = \frac{1}{a(b_n)}$ *where* $b_n = G^{-1}(1 - n^{-1/2})$

Assumption 2.(i) allows one to flexibly control the relative size of each side of the market

through the parameters γ_w and γ_m. Assumption 2.(ii) guarantees that, in each period t, the probability that workers or firms stay unmatched does not become degenerate in the limit. If the size of the outside option does not grow with the size of the market, the probability that it becomes dominated by an alternative option will tend to one given that taste shocks have unbounded support.[5] Assumption 2.(iii) controls the scale of the unobserved shocks such that both the unobserved and systematic parts of the payoffs stay predictive of agents' choices in the limit. Given that U_t and V_t are bounded and that the support of taste shocks is unbounded, U_t and V_t would become irrelevant in the limit without this restriction. More specifically, if G is Gumbel, then we pick $b_n \asymp \frac{1}{2}\log(n)$ and $\sigma_n = 1$.[6] If taste shocks are standard normal, we pick $b_n \asymp \sqrt{\log n}$ and $\sigma_n \asymp b_n$ and for Gamma distributed taste shocks, we pick $b_n \asymp \log(n)$ and $\sigma_n = 1$.

2.4 Equilibrium

I now specify the equilibrium concept mapping agents' preferences to the observed matching μ. Pairwise stability is a standard solution concept in the two-sided matching literature (Roth and Sotomayor, 1992). I introduce this notion in the following definition.

Definition 1 *A matching μ is stable in period t if and only if for all $i = 1, ..., n_w$ and $j = 1, ..., n_m$:*
(i) Individual rationality: $U_{i\mu_w(i,t)t} \geq U_{i0t}$ and $V_{\mu_m(j,t)jt} \geq V_{0jt}$.
(ii) No blocking pairs: There exists no pair (i, j) such that $U_{ijt} > U_{i\mu_w(i,t)t}$ and $V_{ijt} > V_{\mu_m(j,t)jt}$.

In a given time period t, a match is stable if and only if agents weakly prefer their matching outcome to the outside option and if there exist no worker-firm pairs that would prefer to rematch together. The set of stable matches corresponds to the core defined by weak domination and is thus a subset of the core. Existence of a stable match and of the worker-optimal/firm-optimal stable matches is guaranteed when preferences are strict and matching is one-to-one (Roth and Sotomayor, 1992).[7] When firms' preferences are heterogeneous,

[5] It will be established later that the size of opportunity sets (defined in Section 2.5) in equilibrium grows at a rate $n^{1/2}$.

[6] The symbol $\asymp$ denotes asymptotic equivalence as n grows to infinity.

[7] The worker-optimal stable match is the most preferred stable outcome from the workers' perspective and the least preferred stable outcome from the firms' perspective. On the contrary, the firm-optimal stable match is the most preferred stable outcome from the firms' perspective and the least preferred stable outcome from the workers' perspective.

many stable matches can exist and the size of the core grows with the size of the market. For an arbitrary vector of continuation values $(\bar{U}, \bar{V})$, I define $\mu[\bar{U}, \bar{V}]$ as the set of matches that are stable in every period $t = 1, ..., T$. I then impose the following assumptions on μ.

Assumption 3 *The realized matching μ satisfies the following conditions:*
(i) Pairwise stability: $\mu \in \mu[\bar{U}, \bar{V}]$.
(ii) Rational expectations: $(\bar{U}, \bar{V}) \in \{(\bar{U}(\mu), \bar{V}(\mu)) : \mu \in \mu[\bar{U}, \bar{V}]\}$.

Assumption 3 imposes that the realized matching μ is stable in each period $t = 1, ..., T$ with respect to agents' lifetime utility. It also imposes that agents have rational expectations and believe that the realized match in future periods will be stable when forming their expectations about the continuation value of a match. Without further restrictions, characterizing the set of equilibrium matches is challenging. In particular, multiplicity in the set of possible realized stable matches in each period generates multiplicity in agents' continuation values. Prior work studying dynamically stable matching has either restricted preferences to ensure uniqueness of the stable outcome in each period (Baccara and Yariv, 2021) or has imposed restrictions on agents' beliefs about the selection mechanism of future stable matches (Doval, 2022). I do not impose such restrictions.

2.5 Opportunity Sets and Inclusive Values

The equilibrium notion described above allows us to express a realized match μ as the solution to individual choice problems. Given an arbitrary realized match μ, we can define the *opportunity set* (or *feasible set*) of a worker in period t as the set of firms that would be willing to hire them instead of their current matched employee in the same period. Similarly, the opportunity set of a firm is the set of workers that would be willing to quit their current employer to work there. Formally, I define the opportunity set faced by a given worker $i \in \mathcal{I}$ in period t given a match μ and a vector of firm continuation values $\bar{V}$ as:

$$M_{it}[\mu, \bar{V}] = \{j \in \mathcal{J} : V_{ijt} \geq V_{\mu_m(j,t)jt}\}.$$

Similarly, I define the opportunity set of firm $j \in \mathcal{J}$ in period t given a match μ and a vector of worker continuation values $\bar{U}$ as:

$$W_{jt}[\mu, \bar{U}] = \{i \in \mathcal{I} : U_{ijt} \geq U_{i\mu_w(i,t)t}\}.$$

The following result shows how one can leverage the notion of opportunity sets to rewrite the conditions imposed by Assumption 3:

Proposition 1 *(i). $\mu \in \mu[\bar{U}, \bar{V}]$ if and only if*

$$U_{i\mu_w(i,t)t} = \max_{k \in M_{it}[\mu,\bar{V}] \cup \{0\}} U_t(x_{it}, z_{kt}) + \sigma\eta_{ikt} + \beta\bar{U}_{ikt}$$

$$V_{\mu_m(j,t)jt} = \max_{l \in W_{jt}[\mu,\bar{U}] \cup \{0\}} V_t(x_{lt}, z_{jt}) + \sigma\epsilon_{ljt} + \delta\bar{V}_{ljt},$$

for all $i = 1, ..., n_w$, $j = 1, ..., n_m$ and $t = 1, ..., T$.
(ii) Under Assumption 1, $(\bar{U}, \bar{V}) \in \{(\bar{U}(\mu), \bar{V}(\mu)) : \mu \in \mu[\bar{U}, \bar{V}]\}$ if and only if

$$\bar{U}_{ijt} \in \left\{ \int_{\mathcal{X}} \mathbb{E}_{(\mu,t)} \left[\max_{k \in M_{it+1}[\mu,\bar{V}] \cup \{0\}} U_{ikt+1} | x_{it+1} \right] w(x_{it+1}|x_{it}, z_{jt}) dx_{it+1} : \mu \in \mu[\bar{U}, \bar{V}] \right\},$$

$$\bar{V}_{ijt} \in \left\{ \int_{\mathcal{Z}} \mathbb{E}_{(\mu,t)} \left[\max_{l \in W_{jt+1}[\mu,\bar{U}] \cup \{0\}} V_{ljt+1} | z_{jt+1} \right] m(z_{jt+1}|x_{it}, z_{jt}) dz_{jt+1} : \mu \in \mu[\bar{U}, \bar{V}] \right\},$$

for all $i = 1, ..., n_w$, $j = 1, ..., n_m$ and $t = 1, ..., T$.

See Appendix A.1 in the Supplementary Material for a proof. Proposition 1 (i) states that a match μ is stable in period t if and only if each worker $i \in \mathcal{I}$ is matched to their preferred firm among their opportunity set and each firm $j \in \mathcal{J}$ is matched to its preferred worker among its opportunity set. Proposition 1 (ii) is a direct implication of (i). This result implies that we can rewrite Assumption 3 as imposing that μ solves a collection of single-agent dynamic discrete choice problems. Characterizing conditional matching probabilities in this problem remains challenging for two reasons. First, agents' realized opportunity sets are unobserved to the econometrician and might be influenced by the realization of their own unobserved preference shocks. Second, agents' continuation values are not uniquely determined due to multiplicity of future realized stable matches.

Section 3 shows how we can circumvent these issues by considering a large market approximation where the number of workers and firms grows to infinity. Our asymptotic argument relies on the fact that, as n grows large, both conditional choice probabilities and continuation values converge to analytical expressions implied by the logit model (McFadden, 1974). This implies that the information contained in opportunity sets necessary to characterize conditional choice probabilities and expected match payoffs can be summarized by sufficient

statistics called *inclusive values*. In anticipation, I define worker i's inclusive value in period t when facing opportunity set M and continuation value $\bar{U}$ as

$$I_{wit}[M, \bar{U}] = n^{-1/2} \sum_{k \in M} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\}.$$

Similarly, I define firm j's inclusive value in period t when facing opportunity set W and continuation value $\bar{V}$ as

$$I_{mjt}[W, \bar{V}] = n^{-1/2} \sum_{l \in W} \exp\left\{V_t(x_{lt}, z_{jt}) + \delta \bar{V}_{ljt}\right\},$$

Note that it will be established later that opportunity sets grow at a rate $n^{1/2}$ under Assumption 1, 2 and 3. Inclusive values are thus normalized to ensure convergence to their conditional expectations as n grows large.[8] Throughout the rest of the paper, I refer to an arbitrary match and vector of continuation values satisfying Assumption 3 as $(\mu^*, \bar{U}^*, \bar{V}^*)$ and denote $I^*_{wit} = I_{wit}[M_{it}[\mu^*, \bar{V}^*], \bar{U}^*]$ and $I^*_{mjt} = I_{mjt}[W_{jt}[\mu^*, \bar{U}^*], \bar{V}^*]$.

3 Asymptotic Characterization

This section leverages the structure described in Section 2 to characterize the limiting matching frequency distributions f_t. I first give the main result in Theorem 1 establishing that, for any $(\mu^*, \bar{U}^*, \bar{V}^*)$ satisfying Assumption 3, the inclusive values I^*_{wit} and I^*_{mjt} converge to the unique solution of a fixed point problem. This implies that the limiting measures f_t are unique despite potential multiplicity in stable equilibria and in agents' continuation values. Moreover, f_t can be expressed as a tractable analytical function of agents' preferences allowing to study the identifying content of data on realized matches. I then provide an outline of the main technical steps of the asymptotic argument.

3.1 Limiting Matching Frequency Distributions

Our main result first establishes that, under Assumption 1, 2, and for any $(\mu^*, \bar{U}^*, \bar{V}^*)$ satisfying Assumption 3, inclusive values have a unique limit and can be approximated by

[8] For convenience, our definition of inclusive values slightly differs from the one used in nested logit models, which refers to the expected utility of the chosen alternative.

solving a fixed point problem. Formally, we state that

$$I^*_{wit} = \Gamma^*_{wt}(x_{it}) + o_p(1) \quad \text{and} \quad I^*_{mjt} = \Gamma^*_{mt}(z_{jt}) + o_p(1), \quad t = 1, ..., T,$$

where the *inclusive value functions* $\Gamma^*_w := (\Gamma^*_{wt})_{t=1}^T$ and $\Gamma^*_m := (\Gamma^*_{mt})_{t=1}^T$ are the unique solutions of the fixed point problem:

$$\Gamma_{wt} = \Psi_{wt}[\Gamma_w, \Gamma_m] \quad \text{and} \quad \Gamma_{mt} = \Psi_{mt}[\Gamma_w, \Gamma_m], \quad t = 1, ..., T. \tag{4}$$

The fixed point mappings Ψ_{wt} and Ψ_{mt} have the following expression

$$\Psi_{wt}[\Gamma_w, \Gamma_m](x) = \int \frac{\exp\left\{U_t(x,s) + V_t(x,s) + \beta \bar{U}_t[\Gamma_w](x,s) + \delta \bar{V}_t[\Gamma_m](x,s)\right\}}{\exp\left\{\delta \bar{V}_t[\Gamma_m](x_{0t}, s)\right\} + \Gamma_{mt}(s)} m_t[\Gamma_w, \Gamma_m](s) ds,$$

$$\Psi_{mt}[\Gamma_w, \Gamma_m](z) = \int \frac{\exp\left\{U_t(s,z) + V_t(s,z) + \beta \bar{U}_t[\Gamma_w](s,z) + \delta \bar{V}_t[\Gamma_m](s,z)\right\}}{\exp\left\{\beta \bar{U}_t[\Gamma_w](s, z_{0t})\right\} + \Gamma_{wt}(s)} w_t[\Gamma_w, \Gamma_m](s) ds,$$

where $\bar{U}_t$ and $\bar{V}_t$ recursively map future inclusive value functions and future continuation values into present continuation values:

$$\bar{U}_t[\Gamma_w](x,z) = \int_{\mathcal{X}} \log\left(\exp\left\{\beta \bar{U}_{t+1}[\Gamma_w](s, z_{0t+1})\right\} + \Gamma_{wt+1}(s)\right) w(s|x,z) ds,$$

$$\bar{V}_t[\Gamma_m](x,z) = \int_{\mathcal{Z}} \log\left(\exp\left\{\delta \bar{V}_{t+1}[\Gamma_m](x_{0t+1}, s)\right\} + \Gamma_{mt+1}(s)\right) m(s|x,z) ds,$$

and m_t and w_t ensure consistency of the aggregate state transitions:

$$w_{t+1}[\Gamma_w, \Gamma_m](x) = \int_{\mathcal{X}} \int_{\mathcal{Z}} w(x|s,h) f_t[\Gamma_w, \Gamma_m](s,h) dh ds + \int_{\mathcal{X}} w(x|s,z_0) f_t[\Gamma_w](s,*) ds,$$

$$m_{t+1}[\Gamma_w, \Gamma_m](z) = \int_{\mathcal{X}} \int_{\mathcal{Z}} m(z|s,h) f_t[\Gamma_w, \Gamma_m](s,h) dh ds + \int_{\mathcal{Z}} m(z|x_0,h) f_t[\Gamma_m](*,h) dh.$$

Our main result then establishes that conditional choice probabilities and continuation values can be approximated by inclusive values in the limit. This implies that, for all $t = 1, ..., T$, and for any μ^* satisfying Assumption 3, there exists a unique limiting measure

$F_t(x,z|\mu^*) = F_t(x,z)$ such that,

$$f_t(x,z) = \frac{\exp\left\{U_t(x,z) + V_t(x,z) + \beta\bar{U}_t^*(x,z) + \delta\bar{V}_t^*(x,z) + \gamma_w + \gamma_m\right\}}{\left(\exp\left\{\beta\bar{U}_t^*(x,z_0)\right\} + \Gamma_{wt}^*(x)\right)\left(\exp\left\{\delta\bar{V}_t^*(x_0,z)\right\} + \Gamma_{mt}^*(z)\right)} w_t^*(x) m_t^*(z),$$

$$f_t(x,*) = \frac{\exp\left\{\beta\bar{U}_t^*(x,z_0) + \gamma_w\right\}}{\exp\left\{\beta\bar{U}_t^*(x,z_0)\right\} + \Gamma_{wt}^*(x)} w_t^*(x),$$

$$f_t(*,z) = \frac{\exp\left\{\delta\bar{V}_t^*(x_0,z) + \gamma_m\right\}}{\exp\left\{\delta\bar{V}_t^*(x_0,z)\right\} + \Gamma_{mt}^*(z)} m_t^*(z), \tag{5}$$

where $\bar{U}_t^* = \bar{U}_t[\Gamma_w^*]$, $\bar{V}_t^* = \bar{V}_t[\Gamma_m^*]$, $w_t^* = w_t[\Gamma_w^*, \Gamma_m^*]$ and $m_t^* = m_t[\Gamma_w^*, \Gamma_m^*]$. This result is formalized in the following theorem:

Theorem 1 *Suppose Assumptions 1 and 2 hold:*
(i) The mapping $(\log\Gamma_w, \log\Gamma_m) \mapsto (\log\Psi_m[\Gamma_w,\Gamma_m], \log\Psi_w[\Gamma_w,\Gamma_m])$ *is a contraction. In particular, the fixed point problem in* (4) *always has a unique solution* Γ_m^*, Γ_w^*.
(ii) For any μ^* *satisfying Assumption 3,* $F_t(x,z|\mu^*) = F_t(x,z)$ *with density* $f_t(x,z)$ *given in* (5).

A proof of Theorem 1 part (i) can be found in Appendix A.3 in the Supplementary Material. Section 3.2 gives an outline of the main steps of the proof of Theorem 1 part (ii). The complete proof, including all intermediary results, can be found in the Supplementary Material. Theorem 1 has several implications. It guarantees existence in the limit of a match μ^* satisfying the equilibrium conditions of Assumption 3. It also implies that all matches satisfying Assumption 3 are observationally equivalent in the limit. Finally, it implies that f_t can be expressed analytically as a tractable function of workers' and firms' payoff functions and inclusive value functions. Inclusive value functions can be derived by iterating a contraction mapping. Table B.1 in the Supplementary Material displays results of Monte Carlo simulations assessing the accuracy of this asymptotic approximation in a simple parametric version of this framework and shows that inclusive values and matching frequencies converge to the theoretical limits defined in (4) and (5).

Equation (5) illustrates the rich empirical content of this framework by providing a clear mapping between workers' and firms' preferences and worker-firm sorting patterns. For example, the degree of matching assortativity between workers of type x and firms of type z in period t could be driven by production complementarities governed by the systematic

part of firms' flow payoffs V_t, or worker preference heterogeneity over z specified by the function U_t, or firms' and workers' continuation value functions. The intensity and direction of worker-firm sorting can also evolve over time in a non-stationary way as preferences change and/or as the distribution of aggregate states evolves endogenously.

Having established a tractable analytical mapping between agents' preferences and observed sorting patterns, Section 4 then investigates whether this mapping is invertible such that preferences could be identified from data on observed matches. Readers primarily interested in the identification results may thus skip Section 3.2 and proceed directly to Section 4.

Remark: Entry and Exit In practice, one might want to relax the assumption that the set of participating agents is fixed over time. I discuss here a simple extension allowing for sequential entry and exit of workers and firms. Assuming that entry and exit is exogenous and occurs at the start of the period, we can rewrite the aggregate state transitions as follows

$$w_{t+1}(x|\mu) = \int_{\mathcal{X}} \int_{\mathcal{Z}} w(x|s,h) dF_{nt}(s,h|\mu) + \int_{\mathcal{X}} w(x|s,z_0) dF_{nt}(s,*|\mu) + w^{\Delta}_{t+1}(x),$$

$$m_{t+1}(z|\mu) = \int_{\mathcal{X}} \int_{\mathcal{Z}} m(z|s,h) dF_{nt}(s,h|\mu) + \int_{\mathcal{Z}} m(z|x_0,s) dF_{nt}(*,s|\mu) + m^{\Delta}_{t+1}(z),$$

where $w^{\Delta}_{t+1}(x)$ and $m^{\Delta}_{t+1}(z)$ denote the net mass increase of workers with observed characteristics x and of firms with observed characteristics z, respectively. I also assume that exit is involuntary and unexpected, meaning that agents always act as if they will be participating in the future. Note that involuntary exit coexists with the option to remain unmatched such that the model allows for both involuntary and voluntary exit.

Under these assumptions, it is immediate to show that Theorem 1 still holds under the following slight modification of the fixed point mappings Ψ_{wt} and Ψ_{mt}:

$$\Psi_{wt}[\Gamma_w, \Gamma_m](x) = \int \frac{\exp\left\{U_t(x,s) + V_t(x,s) + \beta \bar{U}_t[\Gamma_w](x,s) + \delta \bar{V}_t[\Gamma_m](x,s)\right\}}{\exp\left\{\delta \bar{V}_t[\Gamma_m](x_{0t},s)\right\} + \Gamma_{mt}(s)} \tilde{m}_t[\Gamma_w, \Gamma_m](s) ds,$$

$$\Psi_{mt}[\Gamma_w, \Gamma_m](z) = \int \frac{\exp\left\{U_t(s,z) + V_t(s,z) + \beta \bar{U}_t[\Gamma_w](s,z) + \delta \bar{V}_t[\Gamma_m](s,z)\right\}}{\exp\left\{\beta \bar{U}_t[\Gamma_w](s,z_{0t})\right\} + \Gamma_{wt}(s)} \tilde{w}_t[\Gamma_w, \Gamma_m](s) ds,$$

where

$$\tilde{m}_t[\Gamma_w, \Gamma_m](z) = m_t[\Gamma_w, \Gamma_m](z) + m_t^{\Delta}(z), \quad \textit{and} \quad \tilde{w}_t[\Gamma_w, \Gamma_m](x) = w_t[\Gamma_w, \Gamma_m](x) + w_t^{\Delta}(x).$$

As w_t^{Δ} and m_t^{Δ} enter additively in $\tilde{w}$ and $\tilde{m}$ and do not depend on Γ_w and Γ_m, the proof of Theorem 1 under this modification remains virtually unchanged.

3.2 Asymptotic Argument: Outline

The asymptotic argument is divided in three steps. I first condition on an arbitrary vector of continuation values to make the problem static and leverage the asymptotic results of Menzel (2015) to show that the conditional choice probabilities and expected matching utilities implied by stability can be approximated by inclusive values. Second, I introduce dynamics and show that present inclusive values are approximated by expected inclusive value functions that depend on (i) the present and future inclusive values of the opposite side of the market and (ii) own future inclusive values. Finally, I show that inclusive value functions are approximate solutions to the sample analog of the fixed point problem (4). Dynamics magnify the dimensionality and complexity of this fixed point problem relative to the static case studied in Menzel (2015) due to interdependence between inclusive values both within and across time periods. Uniform convergence of the fixed point mappings to Ψ_w and Ψ_m then implies that inclusive values converge to the unique solutions of the population fixed point problem Γ_w^* and Γ_m^*.

Conditional Choice Probabilities and Expected Matching Utility I start by characterizing the limit of conditional choice probabilities and expected match payoffs as the number of alternatives grows to infinity assuming that agents' choice sets and continuation values are observed and exogenous.

Proposition 2 *Suppose Assumption 1 holds. Then as* $J \to \infty$,
(i).

$$\mathbb{P}(U_{i0t} \geq \max_{k=0,\ldots,J} U_{ikt} | x_{it}, (z_{kt})_{k=1}^J, (\bar{U}_{ikt})_{k=0}^J)$$

$$= \frac{\exp\left\{\beta \bar{U}_{i0t}\right\}}{\exp\left\{\beta \bar{U}_{i0t}\right\} + \frac{1}{J}\sum_{k=1}^{J} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\}} + o(1),$$

$$J\mathbb{P}(U_{ijt} \geq \max_{k=0,...,J} U_{ikt} | x_{it}, (z_{kt})_{k=1}^{J}, (\bar{U}_{ikt})_{k=0}^{J})$$
$$= \frac{\exp\left\{U_t(x_{it}, z_{jt}) + \beta \bar{U}_{ijt}\right\}}{\exp\left\{\beta \bar{U}_{i0t}\right\} + \frac{1}{J}\sum_{k=1}^{J} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\}} + o(1),$$

for any fixed $j = 1, ..., J$.

(ii).

$$\mathbb{E}\left[\max_{k=0,...,J} U_{ikt} | x_{it}, (z_{kt})_{k=1}^{J}, (\bar{U}_{ikt})_{k=0}^{J}\right]$$
$$= \log\left(\exp\left\{\beta \bar{U}_{i0t}\right\} + \frac{1}{J}\sum_{k=1}^{J} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\}\right) + \log J + \gamma + o(1).$$

See Appendix A.2 in the Supplementary Material for a proof of this result. The same result holds symmetrically for the firm side. Proposition 2 shows that, under Assumption 1 and 2, conditional choice probabilities converge to the Logit formula as the number of alternatives grows to infinity. The expected value of the chosen alternative also converges to the well-known log-sum expression (McFadden, 1974). This result is a direct implication of the type-I upper tail assumption and shows that the choice of distribution G within the domain of attraction of the Gumbel distribution has no consequence for conditional choice probabilities and expected match payoffs. To simplify notations, I subtract $\log J + \gamma$ from agents' continuation values WLOG throughout the rest of the analysis, as these terms cancel out in conditional choice probabilities. Note that the conditional choice probability of choosing a particular alternative j would converge to zero if we do not weight it by J. Lemma 1 in the Supplementary Material establishes that the size of opportunity sets increases at a rate $\sqrt{n}$ which justifies Assumption 2.(ii).

In practice, one cannot directly leverage Proposition 2 to characterize the conditional matching probabilities implied by Proposition 1 as agents' opportunity sets are unobserved equilibrium objects. In particular, Assumption 3 implies that opportunity sets are potentially endogenous as shifting worker i's taste shocks could trigger a chain of rematches that could

affect their own opportunity set. Using a similar argument as in Menzel (2015), one can show that, as the size of the market increases, the probability for such an event to occur vanishes to zero. This result stems mostly from two implications of Proposition 2: (i) the probability that firm j rematches with a specific worker i vanishes to zero as the size of the market increases and (ii) the probability of choosing the outside option instead, which would terminate such a chain of rematches, is non degenerate in the limit. This result is formalized in Lemma 2 in the Supplementary Material where a more detailed discussion and proof can be found.

As a result of Lemma 2, we can show that the dependence between taste shocks and opportunity sets vanishes in the limit. This means that the distribution of taste shocks conditional on current and future realizations of opportunity sets converges to their marginal distribution g. Again, this result is formalized in Lemma 3 in the Supplementary Material. Note that these conditional distributions are only well defined for the extremal matchings, as they are the only stable matchings that always exist irrespective of the size of the market. We can thus use Proposition 2 and inclusive values to approximate conditional choice probabilities and expected matching utilities that would arise under the extremal matchings. The rest of the argument consists in characterizing inclusive values' behavior in the limit.

Expected Inclusive Value Functions Conditional on an arbitrary vector of continuation values $(\bar{U}, \bar{V})$, i.e. abstracting away from dynamics for now, inclusive values arising from the extremal matchings are approximated by *expected inclusive value functions*. We can define $M_{it}^M[\bar{U}, \bar{V}]$ and $M_{it}^W[\bar{U}, \bar{V}]$ as the opportunity set faced by worker i in period t under the firm-optimal and worker-optimal stable match, respectively. We can define analogously $W_{jt}^M[\bar{U}, \bar{V}]$ and $W_{jt}^W[\bar{U}, \bar{V}]$ for firm j in period t. From there, we can rewrite $I_{wit}^M[\bar{U}, \bar{V}] := I_{wit}[M_{it}^M[\bar{U}, \bar{V}], \bar{V}]$ as follows

$$\begin{aligned} I_{wit}^M[\bar{U}, \bar{V}] &= \frac{1}{n} \sum_{k=1}^{n_m} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt})\right\} \times \sqrt{n} \mathbb{1}\{k \in M_{it}^M[\bar{U}, \bar{V}]\} \\ &= \frac{1}{n} \sum_{k=1}^{n_m} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt})\right\} \times \sqrt{n} \mathbb{1}\{V_{ikt} \geq \max_{l \in W_{kt}^M[\bar{U}, \bar{V}] \cup \{0\}} V_{lkt}\}. \end{aligned}$$

Worker i's inclusive value is thus a weighted average of the probability that a firm belongs to their opportunity set, which itself depends on firms' preferences and opportunity sets. We

can use Lemma 3 and Proposition 2 (i) to show that

$$I^M_{wit}[\bar{U},\bar{V}] \geq \hat{\Gamma}^M_{wt}[\bar{U},\bar{V}](x_{it}) + o_p(1) \quad \text{and} \quad I^M_{mjt}[\bar{U},\bar{V}] \leq \hat{\Gamma}^M_{mt}[\bar{U},\bar{V}](z_{jt}) + o_p(1),$$

where $\hat{\Gamma}^M_{wt}$ and $\hat{\Gamma}^M_{mt}$ are the firm-optimal *expected inclusive value functions* of workers and firms, respectively, defined as

$$\hat{\Gamma}^M_{wt}[\bar{U},\bar{V}](x_{it}) = \frac{1}{n}\sum_{k=1}^{n_m} \frac{\exp\left\{U_t(x_{it},z_{kt}) + V_t(x_{it},z_{kt}) + \beta\bar{U}_{ikt} + \delta\bar{V}_{ikt}\right\}}{\exp\left\{\delta\bar{V}_{0kt}\right\} + I^M_{mkt}[\bar{U},\bar{V}]},$$

$$\hat{\Gamma}^M_{mt}[\bar{U},\bar{V}](z_{jt}) = \frac{1}{n}\sum_{l=1}^{n_w} \frac{\exp\left\{U_t(x_{lt},z_{jt}) + V_t(x_{lt},z_{jt}) + \beta\bar{U}_{ljt} + \delta\bar{V}_{ljt}\right\}}{\exp\left\{\beta\bar{U}_{l0t}\right\} + I^M_{wlt}[\bar{U},\bar{V}]}.$$

Similar bounds can be established for the inclusive values that would arise under the worker-optimal stable match with the direction of the inequalities reversed. A formal exposition and proof of this result can be found in Lemma 4 in the Supplementary Material.

We can then introduce dynamics by leveraging the fact that, conditional on an arbitrary vector of continuation values $(\bar{U},\bar{V})$, expected matching utilities arising from the extremal matchings can be approximated by expected inclusive value functions. We define $\bar{U}^M_{ijt}[\bar{U},\bar{V}]$ as the expected future matching lifetime utility of worker i conditional on matching with firm j in period t, under the sequence of firm-optimal stable matches. Using Proposition 2 (ii) and Lemma 4 we can show that,

$$\begin{aligned}\bar{U}^M_{ijt}[\bar{U},\bar{V}] &\geq \int \log\left(\exp\left\{\beta\bar{U}^M_{i0t+1}[\bar{U},\bar{V}]\right\} + I^M_{wit+1}[\bar{U},\bar{V}]\right) w(x_{it+1}|x_{it},z_{jt})dx_{it+1} + o_p(1) \\ &\geq \int \log\left(\exp\left\{\beta\bar{U}^M_{i0t+1}[\bar{U},\bar{V}]\right\} + \hat{\Gamma}^M_{wt+1}[\bar{U},\bar{V}](x_{it+1})\right) w(x_{it+1}|x_{it},z_{jt})dx_{it+1} + o_p(1).\end{aligned}$$

Similarly, we can show that

$$\bar{V}^M_{ijt}[\bar{U},\bar{V}] \leq \int \log\left(\exp\left\{\beta\bar{V}^M_{0jt+1}[\bar{U},\bar{V}]\right\} + \hat{\Gamma}^M_{mt+1}[\bar{U},\bar{V}](z_{jt+1})\right) m(z_{jt+1}|x_{it},z_{jt})dz_{jt+1} + o_p(1),$$

where the sign the inequalities are reversed for a sequence of worker-optimal stable matches.

Fixed Point Representation In sum, conditional on agents' continuation values, Assumption 3 (i) implies that the inclusive value of a given worker can be approximated by a

function of firms' preferences and inclusive values. Assumption 3 (ii) further implies that the continuation values of workers and firms can be approximated by their own future inclusive values. Hence, the two-sided and dynamic nature of the problem implied by Assumption 3 gives rise to a fixed point problem characterizing these inclusive values within and across time periods. I define the fixed point mappings as

$$\hat{\Psi}_{wt}[\Gamma_w, \Gamma_m](x) = \frac{1}{n}\sum_{k=1}^{n_m} \frac{\exp\left\{U_t(x, z_{kt}) + V_t(x, z_{kt}) + \beta \bar{U}_t[\Gamma_w](x, z_{kt}) + \delta \bar{V}_t[\Gamma_m](x, z_{kt})\right\}}{\exp\left\{\delta \bar{V}_t[\Gamma_m](x_{0t}, z_{kt})\right\} + \Gamma_{mt}(z_{kt})},$$

$$\hat{\Psi}_{mt}[\Gamma_w, \Gamma_m](z) = \frac{1}{n}\sum_{l=1}^{n_w} \frac{\exp\left\{U_t(x_{lt}, z) + V_t(x_{lt}, z) + \beta \bar{U}_t[\Gamma_w](x_{lt}, z) + \delta \bar{V}_t[\Gamma_m](x_{lt}, z)\right\}}{\exp\left\{\beta \bar{U}_t[\Gamma_w](x_{lt}, z_{0t})\right\} + \Gamma_{wt}(x_{lt})},$$

where

$$\bar{U}_t[\Gamma_w](x, z) = \int_{\mathcal{X}} \log\left(\exp\left\{\beta \bar{U}_{t+1}[\Gamma_w](s, z_{0t+1})\right\} + \Gamma_{wt+1}(s)\right) w(s|x, z) ds,$$

$$\bar{V}_t[\Gamma_m](x, z) = \int_{\mathcal{Z}} \log\left(\exp\left\{\delta \bar{V}_{t+1}[\Gamma_m](x_{0t+1}, s)\right\} + \Gamma_{mt+1}(s)\right) m(s|x, z) ds.$$

The proof of Theorem 1 part (ii) in the Supplementary Material then leverages Lemma 4 and the fact that $\hat{\Psi}_{wt}[\Gamma_w, \Gamma_m]$ and $\hat{\Psi}_{mt}[\Gamma_w, \Gamma_m]$ are nonincreasing in Γ_m and Γ_w, respectively, nondecreasing in Γ_w and Γ_m, respectively, and Lipschitz continuous in (Γ_w, Γ_m) to show that for any collection of functions $(\Gamma^*_{w1}, ..., \Gamma^*_{wT})$ and $(\Gamma^*_{m1}, ..., \Gamma^*_{mT})$ solving the fixed-point problem

$$\Gamma^*_{wt} = \hat{\Psi}_{wt}[\Gamma^*_w, \Gamma^*_m] + o_p(1) \quad \text{and} \quad \Gamma^*_{mt} = \hat{\Psi}_{mt}[\Gamma^*_w, \Gamma^*_m] + o_p(1), \quad t = 1, .., T,$$

we have for any $(\mu^*, \bar{U}^*, \bar{V}^*)$ satisfying Assumption 3

$$\hat{\Gamma}^*_{wt} = \Gamma^*_{wt} + o_p(1) \quad \text{and} \quad \hat{\Gamma}^*_{mt} = \Gamma^*_{mt} + o_p(1),$$

Lemma 5 in the Supplementary Material establishes uniform convergence of the finite sample fixed point mappings $\hat{\Psi}_w$ and $\hat{\Psi}_m$ to their population counterparts Ψ_w and Ψ_m which implies uniform convergence of $\hat{\Gamma}$ to Γ^* the solution of the population fixed-point problem (4). This completes the proof of convergence of inclusive values to Γ^*. We then use Proposition 1 and 2 to derive the limit of conditional matching probabilities and finish proving Theorem 1.

4 Identification and Estimation

This section leverages the asymptotic characterization of the matching frequency distributions f_t described in Section 3 to investigate whether workers' and firms' preferences can be identified and estimated from data on realized matches.

4.1 Sampling Process

The limiting joint density f_t can be identified from data on observed matches under various sampling schemes. One can sample individuals at random regardless of whether they are firms or workers. One observation in a given period t is thus composed of this individual alone, if they are unmatched, or along with their matched partner otherwise. The probability that a matched individual is selected by this sampling process is thus twice the probability that an unmatched individual is selected. The joint density function of matched characteristics h_t arising from this sampling process relates to f_t in the following way:

$$h_t(x,z) = \frac{2f_t(x,z)}{\exp\{\gamma_w\} + \exp\{\gamma_m\}},$$

where $h_t(x,z)$ is the mass of firms with observed characteristics z matched with workers with observed characteristics x in period t arising from the sampling scheme defined above and $\exp\{\gamma_w\} + \exp\{\gamma_m\}$ is the total mass of workers and firms available in this economy. Similarly, I define:

$$h_t(x,*) = \frac{f_t(x,*)}{\exp\{\gamma_w\} + \exp\{\gamma_m\}},$$

$$h_t(*,z) = \frac{f_t(*,z)}{\exp\{\gamma_w\} + \exp\{\gamma_m\}},$$

where $h_t(*,z)$ is the mass of unmatched firms with observed characteristics z and $h_t(x,*)$ is the mass of unmatched workers.

The aggregate distributions of observed attributes m_t and w_t can be directly identified from f_t as follows:

$$\int_{\mathcal{Z}_t} f_t(x,z)dz + f_t(x,*) = w_t(x)\exp\{\gamma_w\},$$

$$\int_{\mathcal{X}_t} f_t(x,z)dx + f_t(*,z) = m_t(z)\exp\{\gamma_m\}.$$

Finally, I assume that we have panel data on observed state transitions such that the Markov

transition probability distributions m and w are nonparametrically identified.

4.2 Identification

We are interested in recovering the payoff functions U_t and V_t for each period t from the functions f_t. Discount factors in dynamic decision processes are typically not separately identified from flow payoff functions (Magnac and Thesmar, 2002). I thus assume that β and δ are known or fixed ex-ante by the econometrician.

I start by studying identification when the sampling scheme allows the analyst to identify f_t for all time periods $t = 1, ..., T$. As $\bar{U}_T = 0$ and $\bar{V}_T = 0$, constructive identification can be established by backward induction. Starting from the last period T, we can identify the *pseudo-surplus function* W_T as follows:

$$W_T(x, z) := U_T(x, z) + V_T(x, z) = \log\left(\frac{f_T(x, z)}{f_T(x, *)f_T(*, z)}\right).$$

We can also identify Γ^*_{wT} and Γ^*_{mT} from the distribution of unmatched workers and firms:

$$\Gamma^*_{wT}(x) = \frac{w_T(x)\exp(\gamma_w)}{f_T(x, *)} - 1,$$

$$\Gamma^*_{mT}(z) = \frac{m_T(z)\exp(\gamma_m)}{f_T(*, z)} - 1.$$

$\bar{U}_{T-1}$ and $\bar{V}_{T-1}$ can then be recovered by backward induction:

$$\bar{U}_{T-1}(x, z) = \int_{\mathcal{X}} \log(1 + \Gamma^*_{wT}(s))w(s|x, z),$$

$$\bar{V}_{T-1}(x, z) = \int_{\mathcal{Z}} \log(1 + \Gamma^*_{mT}(s))m(s|x, z).$$

From there, one can iterate backward to identify the inclusive value functions and the surplus function in period $T - 1$ and all the preceding periods. However, it is immediate to see from (5) that the following transformation of agents' payoff functions $\tilde{U}_t = aU_t$ and $\tilde{V}_t = V_t + (1 - a)U_t$ would leave f_t unchanged for any real finite number a. This implies that agents' individual preferences are not separately identified from data on observed matches without further assumptions. This result is formalized in the following proposition.

Proposition 3 *Under Assumption 1-3:*
(i) The pseudo-surplus function W_t and the inclusive value functions Γ^*_{wt} and Γ^*_{mt} solving (4) are identified for $t = 1, ..., T$ from the limiting joint distributions of matched characteristics $f_1, ...f_T$ defined in (5).
(ii) Without further restrictions, U_t and V_t are not identified for any $t = 1, ..., T$ from the limiting joint distributions of matched characteristics $f_1, ...f_T$ defined in (5).

While it appears restrictive that individual preferences cannot be identified from data on realized matches, one can generate rich counterfactual predictions about the distribution of matched characteristics from the surplus function and inclusive value functions only. Specifically, any change in the distribution of initial observed attributes w_1 and m_1, the transition distribution functions w and m, the pseudo-surplus functions W_t or the discount factors β and δ can be mapped into the corresponding counterfactual matching frequency distributions f_t as well as the counterfactual aggregate state distributions m_t and w_t. Counterfactual ex-ante expected match payoffs can also be derived from inclusive values allowing to make welfare assessments without needing to know individual preferences.

Separately identifying preferences might be necessary to evaluate other counterfactuals. For example, simulating the effect of changes in income tax policy on sorting would require identifying labor supply elasticities. We further show that any additional data or restriction that allows one to observe the opportunity sets (or a random subset of opportunity sets) of agents on one side of the market is sufficient to separately identify preferences from the surplus. Assume that, in each period, we observe an exogenous subset of firms' opportunity sets W^*_{jt}. These additional data directly identify $s_{wt}(x|z)$ defined as the limiting distribution of x_{it} conditional on the event $i \in W^*_{jt}$ and $z_{jt} = z$. Proposition 2 and Theorem 1 directly imply that under Assumptions 1, 2 and 3:

$$s_{wt}(x|z) = \frac{\exp\left\{U_t(x,z) + \beta \bar{U}^*_t(x,z) + \gamma_w\right\}}{\exp\left\{\beta \bar{U}^*_t(x,z_0)\right\} + \Gamma^*_{wt}(x)} w(x). \tag{6}$$

One can then proceed by backward induction and identify V_T and U_T:

$$V_T(x,z) = \log \frac{f_T(x,z)}{s_{wT}(x|z) f_T(*,z)}, \quad \textit{and} \quad U_T(x,z) = \log \frac{s_{wT}(x|z)}{f_T(x,*)},$$

and iterate backward to identify U_t and V_t for $t = 1, ..., T$. This result is formalized in the

following proposition.

Proposition 4 *Under Assumption 1-3, the payoff functions U_t and V_t are identified for $t = 1, ..., T$ from the limiting joint distributions of matched characteristics $f_1, ...f_T$ defined in (5) and the limiting conditional distributions of worker characteristics belonging to firms' opportunity sets $s_{w1}, ..., s_{wT}$ defined in (6).*

Note that the same argument applies to the symmetric case where one would instead observe an exogenous subset of workers' opportunity sets. I leverage this result in the empirical application in Section 6 by using additional data on a set of candidates that are exogenously shortlisted to be interviewed by schools to separately identify teachers and schools preferences.

Remark: Identification with Short Panels In practice, one might be interested in studying identification when our sampling scheme identifies the matching frequency distributions f_t for a subset of periods only. I introduce the following assumption.

Assumption 4 *Preferences and aggregate state distributions are on a stationary path: $U_t = U$, $V_t = V$, $\bar{U}_t^* = \bar{U}^*$, $\bar{V}_t^* = \bar{V}^*$, $m_t^* = m_1$ and $w_t^* = w_1$ for $t = 1, ..., T$.*

Under Assumption 1, 2, 3 and 4, inclusive value functions are also stationary $\Gamma_{mt}^* = \Gamma_m^*$, $\Gamma_{wt}^* = \Gamma_w^*$ for $t = 1, ...T$. In practice, existence of such a stationary equilibrium might entail agents exiting and entering the market to ensure stationarity of the distribution of aggregate states while allowing individual states to evolve over time. Section 3.1 discusses how Theorem 1 extends to a setting allowing for exogenous entry and exit. Under Assumption 4, we can use (5) to show that $\bar{U}^*$ solves the following fixed point problem:

$$\bar{U}^*(x,z) = h[\bar{U}^*](x,z) = \beta \int \bar{U}^*(s,z_0) w(s|x,z) ds - \int \log \frac{f(s,*)}{w^*(s)\exp(\gamma_w)} w(s|x,z).$$

As $\beta < 1$, it is immediate to show that h is a contraction mapping such that there exists a unique solution $\bar{U}^*$ to this fixed point problem. This argument allows us to identify the continuation value functions $\bar{U}^*$ and $\bar{V}^*$ from any f_t. From there, we can identify the pseudo-surplus function

$$W(x,z) = \log\left(\frac{f_t(x,z)\exp\{\beta\bar{U}^*(x,z_0) + \delta\bar{V}^*(x_0,z)\}}{f_t(x,*)f_t(*,z)\exp\{\beta\bar{U}^*(x,z) + \delta\bar{V}^*(x,z)\}}\right).$$

As in the non-stationary case, U and V are not separately identified without further restrictions. However, additional data on the opportunity sets of one side of the market are sufficient to recover separate identification of U and V:

$$U(x,z) = \log\left(\frac{f_t(x,z)\exp\{\delta\bar{V}^*(x_0,z)\}}{s_{wt}(x|z)f_t(*,z)\exp\{\delta\bar{V}^*(x,z)\}}\right),$$

$$V(x,z) = \log\left(\frac{s_{wt}(x|z)\exp\{\beta\bar{U}^*(x_0,z)\}}{f_t(x,*)\exp\{\beta\bar{U}^*(x,z)\}}\right).$$

This result is formalized in the following proposition.

Proposition 5 *Under Assumption 1-4:*
(i) The pseudo-surplus function W and the inclusive value functions Γ_w^* and Γ_m^* solving (4) are identified from any f_t defined in (5).
(ii) Without further restrictions, U and V are not identified from any f_t defined in (5).
(iii). The payoff functions U and V are identified from any (f_t, s_{wt}) defined in (6) and (5).

Note that in the stationary case, a single cross section is sufficient to identify and estimate W only if the transition distribution functions w and m are identified. In general, it is thus necessary to observe f_t for at least two consecutive periods.

4.3 Estimation

I consider a parametric version of this framework where I define the payoff functions as $U(x,z;\theta_t)$ and $V(x,z;\theta_t)$ such that U and V are known up to a vector of parameters θ_t. I assume that we observe a random sample of K individuals in each period t drawn from the sampling scheme described in Section 4.1, along with their respective matches. For a given observation k in period t, we observe a vector $(x_t(k), z_t(k))$ which is encoded differently depending on the type of match we observe. For an unmatched worker, indexed by $w_t(k) = 0$, I record their characteristics in $x_t(k)$ and encode $z_t(k)$ as missing. Similarly, for an unmatched firm, indexed by $m_t(k) = 0$, I record its characteristics in $z_t(k)$ and encode $x_t(k)$ as missing. For a matched worker or firm, indexed by $m_t(k) = w_t(k) = 1$, I record their characteristics along with the characteristics of their match in $(x_t(k), z_t(k))$. We can then construct the

following sample average log-likelihood:

$$L(\boldsymbol{\theta}) = \frac{1}{KT}\sum_{t=1}^{T}\sum_{k=1}^{K}\log\big[\mathbb{1}\{w_t(k)=0\}h_t(x_t(k),*,\theta_t) + \mathbb{1}\{m_t(k)=0\}h_t(*,z_t(k),\theta_t) + \mathbb{1}\{m_t(k)=1,w_t(k)=1\}h_t(x_t(k),z_t(k),\theta_t)\big]. \quad (7)$$

Deriving the log-likelihood function for a given parameter vector $\boldsymbol{\theta}$ involves solving the finite sample version of the fixed point problem (4) to derive the inclusive values $\Gamma^*_{wt}(x_t(k))$, $\Gamma^*_{mt}(z_t(k))$ and the continuation values $\bar{U}^*_t(x_t(k),z_t(k))$, $\bar{V}^*_t(x_t(k),z_t(k))$ for each individual k and period t. This can be achieved in practice by applying the contraction mappings $\hat{\Psi}$ until convergence. The estimator proposed is then defined as:

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}\in\Theta} L(\boldsymbol{\theta})$$

Table B.2 in the Supplementary Material displays results of Monte Carlo simulations assessing the performance of this estimation strategy in a simple parametric version of this framework and shows that it delivers unbiased and consistent estimates of $\boldsymbol{\theta}$. Asymptotic inference for $\hat{\boldsymbol{\theta}}$ is standard if the sample used for estimation is not too large relative to the overall size of the market. As noted in Menzel (2015), inference from a sample including all agents from a market would need to account for dependence of matching outcomes across individuals. Diamond and Agarwal (2017) develop a bootstrap procedure accounting for such dependence in a static two-sided matching model with NTU and homogeneous preferences on both sides of the market. Similar tools could be expanded to a broader class of matching models, including the one presented in this paper, and is left for future work.

Remark: Time-invariant Unobserved Heterogeneity Accounting for unobserved preference heterogeneity in dynamic models is crucial to explain observed behavior. In particular, time-invariant unobserved preference heterogeneity allows to explain persistence in observed choices that cannot be rationalized by moving or switching costs. I consider a setting where $x_{it} = (y'_{it}, \kappa'_i)'$ and $z_{jt} = (v'_{jt}, \xi'_j)'$ where κ_i and ξ_j are vectors of time-invariant worker and firm attributes, respectively, while y_{it} and v_{jt} are observed. I assume that $\kappa_i|y_{i1} \sim P(\kappa)$ and $\xi_j|v_{j1} \sim Q(\xi)$ are conditionally i.i.d. for all $i = 1, ..., n_w$ and $j = 1, ..., n_m$.

We can express the limiting matching density f_t as a function of observables only as

follows

$$\tilde{f}_t(y,v) = \int f_t(x,z)dP(\kappa)dQ(\xi),$$
$$= \int \frac{\exp\left\{U_t(x,z) + V_t(x,z) + \delta\bar{V}_t^*(x,z) + \beta\bar{U}_t^*(x,z)\right\}}{(\exp\left\{\beta\bar{U}_t^*(x,z_0)\right\} + \Gamma_{wt}^*(x))(\exp\left\{\delta\bar{V}_t^*(x_0,z)\right\} + \Gamma_{mt}^*(z))} w_t^*(y)m_t^*(v)dP(\kappa)dQ(\xi)$$

where the inclusive value functions and continuation values solve the fixed point problem (4) conditional on κ and ξ. Identification and estimation of the mixing distributions P and Q typically benefits from the availability of panel data recording repeated matching decisions from the same workers and firms (Kasahara and Shimotsu, 2009). This information is summarized by the joint distribution of matched characteristics across time periods $\int \prod_t f_t(x,z)dP(\kappa)dQ(\xi)$, which can be incorporated in the log-likelihood function.

A commonly used specification in dynamic models discretizes unobserved heterogeneity by indexing payoff functions by these unobserved states and assuming a finite number of support points for their distributions. A similar approach can be used in this setting where the analyst can use the EM algorithm to jointly estimate (i) the conditional probabilities that workers and firms belong to each unobserved "type" and (ii) type-specific surplus functions (Arcidiacono and Miller, 2011). One can also use a two-step grouped fixed-effect approach and classify agents into groups in a first-step, based on moments that are informative about the underlying unobserved heterogeneity, and estimate group-specific surplus functions in a second step (Bonhomme et al., 2022). In the empirical application in Section 6, I use the latter approach to capture persistent unobserved teacher preferences for urban vs. rural areas.

In the rest of the paper, I apply the identification and estimation strategy presented in this section to study the dynamics of the teacher labor market. I leverage rich panel data on the centralized recruitment drive for public teachers in Peru to estimate teachers' and schools' preferences and evaluate counterfactual policies aimed at reducing urban-rural disparities in student achievement caused by teacher attrition. I first provide a brief description of key features of the institutional context and the available data and provide descriptive evidence on the role of teacher attrition in explaining geographical disparities in student achievement. I then discuss the empirical strategy before showing the results of the analysis.

5 Data and Descriptive Analysis

5.1 Context and Data

Public teachers in Peru are hired under permanent or temporary contracts. All temporary contracts have a fixed term of two years and offer the same baseline salary. Permanent contracts do not have a fixed term and offer a baseline salary that scales with experience.[9] On top of the baseline salary, S/1550 (peruvian Soles) in 2015, bonuses are attributed to teachers working in particular schools and areas. Specifically, schools with classrooms regrouping several grades (*multigrado*) provide a monthly wage bonus of S/140, schools with a single teacher (*unidocente*) provide a bonus of S/200, schools located in zones with a high drug cartel presence (*VRAEM*) provide a bonus of S/300, schools located close to the country borders (*frontera*) provide a bonus of S/100 and schools teaching in several languages (*bilingue*) provide a bonus of S/50. Finally, a set of monthly wage bonuses ranging from S/70 to S/500 that scale with how remote a school is are attributed based on cutoff rules on the population of the schools' locality and the travel distance between the provincial capital and the school.

Since 2015, all new temporary and permanent positions are allocated through a centralized recruitment drive than runs between November and March every two years. Teachers already holding a permanent contract cannot participate in this process and need to go through a separate decentralized procedure if they wish to be transferred to another school. The national recruitment drive is divided in three steps:

- *National Competency Test*: All applicants to the recruitment drive have to take a standardized competency test. The test has three components: reading comprehension, logical reasoning and curricular knowledge. To be eligible for a permanent position, applicants need a score of at least 60% in each component of the test. Only 9% of applicants meet this requirement (see Table C.1).

- *Allocation of permanent positions*: The Ministry of Education makes the list of available permanent positions publicly available. Applicants eligible for a permanent po-

[9]The pay scale is divided in six categories. Teachers with a permanent contract can apply once a year for a promotion. Promotions are awarded through a national standardized evaluation and a decentralized evaluation from a local committee. The baseline salary at the highest scale is 75% higher than the lowest scale.

sition choose a province and submit a rank order list of schools within this province. Each school then receives a shortlist of up to ten applicants that they evaluate in a decentralized stage involving an in-person interview. These shortlists are formed using the following variant of serial dictatorship: applicants are ranked according to their test scores and are assigned sequentially, by order of their rank, to their three top ranked schools, among those that still have available interview slots. Once schools have evaluated all applicants in their shortlist, they rank candidates and make offers sequentially according to this ranking. All unassigned applicants can then participate in an aftermarket allocating the remaining unfilled permanent positions or participate in the allocation of temporary positions which takes place shortly after.

- *Allocation of temporary positions*: All participating applicants choose a province. Within each province, applicants are assigned to schools through serial dictatorship using test scores as priorities. Schools do not have any role in the allocation process and cannot express their preferences over applicants. All unassigned applicants can participate in an aftermarket allocating the remaining unfilled temporary positions.

I combine several administrative datasets to create a comprehensive record of all applicants and job postings within the centralized application platform over the 2015-2019 period. The *centralized assignment data* collects information on all stages of the recruitment drive in each year. I observe applicants' scores in the national competency test along with other demographics, the set of posted permanent and temporary positions, the set of shortlisted candidates for permanent positions in each school, and the final allocation of both temporary and permanent positions. I am able to track applicants across years to construct a panel of teachers' re-matching decisions. I complement this dataset with several auxiliary sources providing additional school and applicant characteristics. The *school census*, collecting information on school infrastructure, local amenities and the GPS coordinates of each school. The *MoE's administrative records*, tracking all employed public-sector teachers from 2012 to 2019 and carrying information on their career trajectories. A government-collected *household-level dataset* primarily used for targeting social programs including additional household characteristics such as their residential location. Finally, a *student-level dataset* recording students' demographics and test scores from the national standardized evaluation in math and Spanish of second and fourth grade students. This dataset records the teacher

identifier corresponding to each classroom and can thus be linked to the centralized assignment dataset. Summary statistics about the set of applicants and positions can be found in Table C.2.

Throughout the rest of the analysis, I use two measures of teacher effectiveness. First, the score teachers get at the national competency test. Second, estimates of teacher value-added derived from the matched teacher-student dataset. I denote Y_{li} the potential value of student l's 4th grade math test score if taught by teacher $i \in \{1, ..., N\}$ such that

$$Y_{li} = \alpha_i + X_l'\beta + \epsilon_{li},$$

where $\mathbb{E}[\epsilon_{li}] = 0$ and $\mathbb{E}[X_l\epsilon_{li}] = 0$ by construction. I denote $S_l \in \{1, ..., N\}$ as the realized assignment of student l to teachers and define $Y_l = \sum_i Y_{li}\mathbb{1}\{S_l = i\}$ as the observed realized test score of student l. Following Chetty et al. (2014a), I assume selection on observables such that $\mathbb{E}[Y_{li}|X_l, S_l] = \alpha_i + X_l'\beta$ for all $i = 1, ..., N$, where X_l includes the following covariates at the student level, averaged at the classroom level and averaged at the school level: lagged math test scores (2nd grade), squared lagged test scores, gender, ethnicity and age. X_l also includes additional school characteristics that could capture differences in teaching conditions and learning environment. Under selection on observables, OLS provides unbiased estimates of (α_i, β) denoted as $(\hat{\alpha}_i, \hat{\beta})$. Note that value-added methods relying on selection on observables have been shown to reliably yield forecast unbiased estimates of teacher and school effects in various settings when controlling for lagged student outcomes (see Bacher-Hicks and Koedel (2023) for a review).

To avoid any attenuation bias resulting from plugging-in the noisy estimates $\hat{\alpha}_i$ in schools' preferences over teacher attributes, I instead use the linear shrinkage estimator. Following Walters (2024), I treat $\hat{\alpha}_i$'s as normally distributed with variance estimates denoted as s_i^2 and assume that the α_i's are drawn i.i.d from a common prior $\alpha_i|s_i \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2)$. I construct unbiased estimates of $(\mu_\alpha, \sigma_\alpha^2)$ as $\hat{\mu}_\alpha = \frac{1}{N}\sum_i \hat{\alpha}_i$ and $\hat{\sigma}_\alpha^2 = \frac{1}{N}\sum_i[(\hat{\alpha}_i - \hat{\mu}_\alpha) - s_i^2]$. I then construct estimated posterior means for the teacher value-added coefficients α_i

$$\hat{\alpha}_i^* = \left(\frac{\hat{\sigma}_\alpha^2}{\hat{\sigma}_\alpha^2 + s_i^2}\right)\hat{\alpha}_i + \left(\frac{s_i^2}{\hat{\sigma}_\alpha^2 + s_i^2}\right)\hat{\mu}_\alpha.$$

I use $\hat{\alpha}_i^*$'s as a measure of teacher effectiveness throughout the empirical analysis. Estimates of β and of the distribution of α_i are reported in Table C.3. Consistently with prior evidence,

test scores are positively correlated with value-added but explain little of its overall variance (Rockoff, 2004).

5.2 Descriptive Evidence

Peru is marked by the presence of large geographical disparities in educational achievement. Children enrolled in rural schools perform 0.58σ lower than children enrolled in urban schools in the national standardized math test. Teacher spatial sorting is an important driver of this gap as teachers working in rural areas have been shown to be substantially less effective than teachers working in urban schools (Bobba et al., 2021). I complement this evidence by documenting how teacher attrition and mobility contribute significantly to the observed urban-rural gap in teacher value-added.

Panel A of Table 1 first documents the sorting patterns of teachers entering the profession through the 2015 recruitment drive. We observe a 0.10σ urban-rural gap in teacher value-added, which corresponds to 17% of the overall urban-rural gap in student achievement. While this initial gap is already substantial, I further document that teacher attrition amplifies it. Panel B of Table 1 decomposes the teacher mobility flows observed between the 2015 recruitment drive and the 2017 recruitment drive by place of origin and arrival. 37% of teachers assigned to a rural school in 2015 either move to an urban school or quit altogether in 2017. On the contrary, only 24% of teachers starting in an urban school move to a rural school or quit. More importantly, teachers leaving a rural school to join an urban school have a value-added 0.03σ higher than teachers staying in a rural school. On the contrary, teachers staying in an urban school have a value-added 0.13σ higher than teachers leaving to a rural school and 0.05σ higher than those quitting. Teachers staying or moving to rural areas are thus negatively selected while teachers staying or moving to urban areas are positively selected.

Overall, these attrition patterns generate a net increase in value-added in urban areas and a net decrease in value-added in rural areas. As a result, Panel C of Table 1 shows that the urban-rural gap in teacher value-added increases by 23% to 0.12σ. Aside from the observed selection of teachers staying in rural areas, it is important to note that the overall attrition rate in rural areas (22%) is larger than in urban areas (16%). This imposes an additional burden on rural schools who need to replace their workforce at a higher rate and cannot benefit from the well documented value-added returns to experience for early career

Table 1: Labor Market Dynamics and Spatial Inequalities

	Share	$\hat{\alpha}_i^*$	Test Score
	(1)	(2)	(3)
Panel A: Initial Sorting			
Urban	0.669	0.092	1.378
Rural	0.331	-0.006	0.609
Panel B: Flows			
From Urban			
to Urban	0.762	0.110	1.442
to Rural	0.073	-0.020	1.079
to Quit	0.165	0.060	1.214
From Rural			
to Urban	0.148	0.013	1.032
to Rural	0.630	-0.020	0.630
to Quit	0.222	0.018	0.269
Panel C: Final Sorting			
Urban	0.560	0.101	1.406
Rural	0.257	-0.020	0.715
Quit	0.183	0.043	0.836

NOTES. Panel A describes how teachers entering the profession in 2015 sort across urban and rural schools. Panel B characterizes how these teachers move in and out of rural and urban schools between 2015 and 2017. Panel C shows how this initial cohort sorts in the 2017 recruitment drive.

teachers (Rockoff, 2004).

Equation (5) sheds light on the potential factors explaining why high value-added teachers seem to increasingly sort in urban schools throughout their career. Teachers' preferences over schools could evolve due to changes in the distribution of x, e.g. through learning or due to life cycle considerations making urban jobs more desirable over time and increasing matching assortativity. High value-added teachers' opportunity sets might also evolve differently as they accumulate experience, retake the national test and become more desirable from the schools' perspective. Schools might also acquire more information about applicants' intrinsic ability over time allowing them to better screen high value-added teachers (Rothstein, 2015). The rest of this section leverages the results described in Section 4 and the available data to estimate teachers' and schools' preferences while accounting for the channels described above. I then use the estimated model to evaluate counterfactual policies targeted at reducing the outflow of high value-added teachers from rural to urban areas.

6 Results

6.1 Parameterization

I consider one side of the market as being teachers and the other side as schools. I assume that a school corresponds to a vacancy such that matching is one-to-one.[10] I generate a random sample of matched teachers and schools following the sampling process described in Section 4.1 using data from the recruitment drives in $t = 2015, 2017, 2019$. Instead of assuming that the continuation value of a match in 2019 is zero, I assume that the horizon faced by agents is infinite but impose stationarity of preferences and of the aggregate distribution of observed attributes (see Assumption 4). Note that observed entry and exit in this market justifies imposing stationarity of aggregate state distributions without restricting individual state transitions. Section 3.1 discusses how Theorem 1 extends to a setting with exogenous entry and exit.

Parameterization Payoffs: I parameterize teachers' payoff function as

$$U(x_{it}, \kappa_i, z_{jt}; \theta) = x'_{1it}(\Theta_1 z_{1jt} + \Theta_2 d_{ijt}) + \kappa'_i(\Theta_3 z_{1jt} + \Theta_4 d_{ijt}), \tag{8}$$

where z_{1jt} is a vector of job attributes including an intercept, an indicator for whether the position is permanent or temporary, the offered wage, an amenity index[11], a set of indicators that determine if the school is eligible to various wage bonuses (*multigrado, unidocente, VRAEM, frontera, bilingue*) and polynomials of the log population of the locality and the travel distance (in hours) between the provincial capital and the locality. Geographical proximity is measured by d_{ijt}, a vector of linear splines of the geodesic distance between school j and teacher i's residence. Preferences over job attributes are allowed to vary with x_{1it}, a vector of teacher characteristics including an indicator for whether their previous job was located in a rural area, and their score at the national competency test. I follow Bonhomme et al. (2022) and use a two-step group fixed-effect approach to capture unobserved

[10] This rules out the presence of complementarities across teachers working in the same school. Stable equilibria need not to exist when allowing for such complementarities (Roth and Sotomayor, 1992). Relaxing this assumption would substantially complicate the analysis and is left for future work.

[11] The index is constructed by aggregating via PCA various measures of school infrastructure quality, the average income of local residents and indicators for access to basic amenities such as internet, electricity or water.

time-persistent preferences for urban vs. rural areas. To do so, I first classify teachers into $K = 3$ groups using kmeans clustering from two sets of moments at the teacher level: (i) the average test score obtained at the National Competency Test over the 2015-2019 period and (ii) the proportion of active yearly employment spells spent working in an urban school over the period 2012-2019. These moments capture individual propensities to work in an urban school conditional on test scores, which are informative about an idiosyncratic latent persistent taste for urban areas. I define κ_i as indicators for teacher i's group identity and allow preferences over z_{1j} and d_{ijt} to flexibly interact with them.

I then parameterize schools' payoff function as

$$V(x_{it}, z_{jt}; \theta) = z'_{2jt}\Theta_5 x_{2it}, \tag{9}$$

where x_{2it} is a vector of teacher attributes including their test score, indicators for their level of experience, gender, shrunken estimates of value-added $\hat{\alpha}_i^*$ and an indicator for whether value-added estimates are available or not. I assume that schools do not observe the value-added of teachers with no prior experience and normalize $\hat{\alpha}_i^*$ to 0. I allow schools' preferences over these teacher attributes to differ according to z_{2jt} which is an indicator for whether school j is located in an urban area or not.

I further adapt the model to accommodate the details of the allocation mechanism of permanent and temporary positions. I assume that a match with a permanent position is a terminating action for both teachers and schools as (i) teachers need to wait at least three years before being able to apply for a transfer and (ii) teachers cannot participate again in the centralized allocation mechanism subsequently. I thus set the discount factors β and δ to 0 in U_{ijt} and V_{ijt} when j is a permanent position. To account for the fact that the allocation of temporary positions is done by serial dictatorship using teacher test scores as priorities (see Section 5.1), I specify the utility that a school with temporary position j gets from being matched with teacher i as $V_{ijt} = s_{it}$, where s_{it} is teacher i's test score in period t. This simplifies the problem as we can directly observe which temporary jobs are in teachers' opportunity sets.

Transition processes: Experience evolves deterministically and exogenously by getting incremented by one every year. The competency score s_{it}, which is contained in x_{it}, evolves

stochastically and exogenously. I assume that the transition distribution function of s_{it} is conditionally normal such that:

$$s_{it+1}|x_{it}, z_{\mu^*(i,t)t} \sim \mathcal{N}(x'_{2it}\gamma_x + z'_{2\mu^*(i,t)t}\gamma_z, x'_{2it}\sigma_x^2 + z'_{2\mu^*(i,t)t}\sigma_z^2), \tag{10}$$

I estimate $(\gamma_x, \gamma_z, \sigma_x, \sigma_z)$ by ML and report the estimates in Table C.4. Finally, the indicator for whether teacher i's previous job was in a rural school included in x_{1it} evolves deterministically and endogenously.

Additional data on schools' opportunity sets: I use additional data on the set of shortlisted candidates that schools interview to construct subsets of schools' opportunity sets and separately identify teachers' and schools' preferences for permanent positions from the surplus (see Proposition 5). This requires assuming that this set of interviewees is exogenous, i.e. the distribution of ϵ conditional on the set of interviewees equals the marginal distribution of ϵ. As the set of interviewees in each school is determined solely by teachers' submitted preference lists and test scores s_{it} (see Section 5.1 for details on the assignment procedure), this would fail only if teachers had private information about schools' unobserved preference shocks ϵ and if it affected the formation of their rank order list. Under this assumption, we can construct exogenous subsets of schools' opportunity sets that we denote $\tilde{W}_j^*$. For every school j, $\tilde{W}_j^*$ is composed of all applicants interviewing for that school who did not accept an offer from a competing school. As stated in Proposition 5, this additional information separately identifies V and U. For every school j, I add the log of $s_w(x|z)$ defined in (6) for every applicant in $\tilde{W}_j^*$ as an additional contribution to the log-likelihood in (7).

Discussion of the stability assumption: Assumption 3 is not directly testable but several properties of the allocation mechanism, described in Section 5.1, limit the presence of frictions that could lead to the existence of blocking pairs. First, the allocation of temporary positions is implemented via serial dictatorship by sequentially asking teachers to choose their preferred position by order of their test score rank. This mechanically precludes the existence of blocking pairs. Second, the formation of shortlists in the allocation of permanent positions is also done via serial dictatorship by sequentially assigning teachers to their top three choices by order of their test score rank. Still, two issues might arise: (i) teachers might

Table 2: Preference Estimates: Teachers

	Baseline	× Previous Job Rural	× Test Score (std.)
	(1)	(2)	(3)
Intercept	6.989 (1.355)	0.890 (1.179)	0.054 (0.560)
Wage	1.859 (0.641)	-1.071 (0.577)	0.060 (0.267)
Amenity Index	0.221 (0.082)	-0.079 (0.066)	-0.047 (0.034)
Distance Spline: 0-20km	-0.122 (0.013)	0.017 (0.010)	-0.010 (0.005)
Distance Spline: 20-100km	-0.053 (0.003)	0.001 (0.003)	-0.001 (0.001)
Distance Spline: 100km+	-0.008 (0.001)	-0.001 (0.001)	0.000 (0.000)
Time to Travel	-0.404 (0.134)	0.262 (0.121)	-0.034 (0.051)
Time to Travel^2	-0.004 (0.003)	0.004 (0.003)	0.001 (0.000)
log(Pop)	0.230 (0.171)	-0.008 (0.131)	0.120 (0.062)
log(Pop)^2	-0.064 (0.009)	0.002 (0.006)	-0.005 (0.003)
Time to Travel × log(Pop)	0.075 (0.018)	-0.044 (0.015)	-0.005 (0.008)
Frontera	0.583 (0.293)	-0.067 (0.254)	0.143 (0.108)
VRAEM	-0.816 (0.530)	0.935 (0.509)	0.074 (0.193)
Multigrado	-0.008 (0.265)	0.286 (0.242)	-0.202 (0.112)
Unidocente	-0.094 (0.371)	0.201 (0.342)	-0.129 (0.166)
Bilingue	-0.989 (0.241)	0.470 (0.228)	-0.127 (0.088)
Permanent	27.579 (0.585)	-	-

NOTES. This table displays estimates of the parameters Θ_1 and Θ_2 described in equation (8). Standard errors are in parentheses.

receive no offers and might want to fill one of the remaining unassigned permanent positions and (ii) teachers might anticipate this possibility and try to avoid it by being strategic when forming their rank order lists. To mitigate the first concern, the Ministry implemented an aftermarket such that all unassigned permanent positions and teachers can meet and match in order to minimize justified envy. Modeling potential mismatches generated by the second concern would be challenging as this would entail having access to data on teachers' beliefs about their chances to succeed at the interviews and developing a dynamic model of strategic reporting where preferences of both sides of the market are unknown. This is beyond the scope of the available data and the proposed methodology and is left for future research.

6.2 Estimates

I report in Table 2 estimates of the parameters of teachers' payoff functions. Normalizing these estimates with respect to the wage coefficient allows us to interpret them as teachers' willingness to pay for non-pecuniary job attributes. Consistently with prior evidence,

Table 3: Preference Estimates: Schools

	Baseline	× Urban
	(1)	(2)
Intercept	3.746 (0.040)	-0.611 (0.044)
Male	0.163 (0.013)	0.052 (0.022)
Experience > 5	0.286 (0.013)	-0.087 (0.020)
Experience = 0	-0.334 (0.025)	0.001 (0.031)
Test Score (std.)	1.938 (0.014)	-0.217 (0.019)
$\hat{\alpha}_i^*$	0.014 (0.062)	0.613 (0.092)
Missing $\hat{\alpha}_i^*$	0.013 (0.019)	0.089 (0.027)

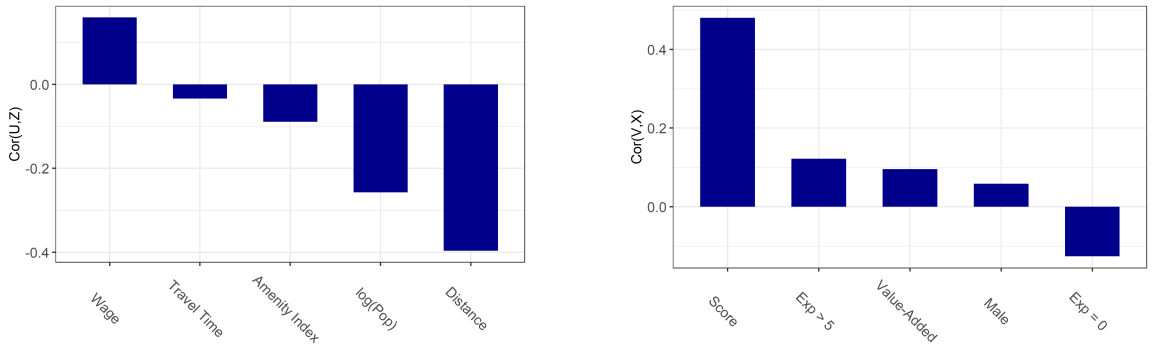
NOTES. This table displays estimates of the parameters Θ_5 described in equation (9). Standard errors are in parentheses.

geographical proximity is an important driver of teachers' preferences. Column (1) shows that increasing the distance between work and home by one kilometer, starting from the 0 to 20 kilometer range, would cost the baseline teacher S/66 monthly (i.e. around 4% of the base salary). This is comparable to the willingness to pay for an increase in the quality of local amenities/infrastructure of 0.55 standard deviations. To directly quantify the relative importance of each job attribute in explaining the overall variation in teachers' preference, I plot in Panel A of Figure 1 the correlation between predicted preferences U_{ijt} for all i, j and t given a random draw of η_{ijt} from the Type-I extreme value distribution and various job attributes. I find that geographical proximity is by far the largest determinant of preferences.

Estimates of preference heterogeneity displayed in columns (2) and (3) suggest that changes in teachers' preferences could play a role in the attrition patterns documented in Table 1. One notable finding is that teachers that were previously working in a rural school have a substantially lower distaste for working in remote areas. This could be indicative of the presence of moving costs or time-persistent unobserved preference heterogeneity for urban vs. rural areas. However, the latter would be at least partly captured by the unobserved types α_i (see Table C.5 for unobserved preference heterogeneity estimates). Similarly, teachers with higher test scores have a larger distaste for low populated areas. As high value-added teachers are more likely to improve their test scores when retaking the exam (see Table C.4), this could explain why they increasingly sort in urban areas throughout their career.

Table 3 displays estimates of the parameters of schools' preferences. I find that schools highly value observed measures of teacher quality such as test scores and experience. Schools

Figure 1: Correlation Preferences



a) Teachers' Preferences

b) Schools' Preferences

Note: Panel a) plots the correlation between teachers' predicted lifetime utility U_{ijt} and various job attributes. Panel b) performs the same exercise and plots the correlation between schools' predicted utility V_{ijt} and various teacher characteristics. Utility is predicted from the estimates reported in Table 2 and 3 and a random draw of taste shocks η_{ijt}, ϵ_{ijt} for all i, j and t.

seem to gender discriminate applicants in favor of males. There is notable heterogeneity in preferences for teacher value-added. While urban schools have a strong preference for high value-added teachers, rural schools put a precise zero weight on it. Panel B of Figure 1 plots the correlation between predicted preferences V_{ijt} for all i, j and t given a random draw of ϵ_{ijt} from the Type-I extreme value distribution and various teacher attributes. Test scores are by far the largest determinant of schools preferences. This suggests that the attrition patterns documented in Table 1 could be driven by changes in teachers' choice sets. As teachers accumulate experience and improve their test score, their priority increases and they become more desirable. As high value-added teachers experience larger test score gains on average than low value-added teachers (see Table C.4), this leads to an outflow of experienced, high-scoring, high value-added teachers from rural to urban areas. As information about teachers' latent value-added gets revealed, urban schools with permanent positions are also more likely to select the highest value-added teachers leading to increased urban-rural sorting in terms of value-added.

Overall, the dynamics of the teacher labor market in Peru seem to exhibit a ladder structure where high value-added teachers starting their career in rural areas become more desirable over time than low value-added teachers and are able to rematch in urban areas. This gives rise to increased sorting of high value-added teachers in urban schools and exacerbates urban-rural disparities in student achievement. This naturally motivates the consideration of policies that would prevent this leakage and retain high value-added teachers in rural

areas, which is the purpose of the next section.

6.3 Counterfactual Policy Evaluation

The estimated model described above allows us to investigate the effectiveness of retention policies aiming at reducing teacher attrition and its adverse effects on inequalities in student achievement. I first evaluate a policy that would mandate a minimum duration for temporary contracts, in similar fashion to what is already implemented for permanent contracts. To do so, I assume that teachers accepting a temporary position in 2015 have to remain in that school and cannot participate in the recruitment drive of 2017 in order to potentially rematch. The lifetime utility that teacher i gets from a temporary position j thus becomes

$$\tilde{U}_{ijt} = U(x_{it}, z_{jt}; \theta) + \eta_{ijt} + \beta \int_{\mathcal{X}} \int_{\mathcal{Z}} (U(s, h; \theta) + \gamma + \beta \bar{U}^*(s, h)) m(h|x_{it}, z_{jt}) w(s|x_{it}, z_{jt}) dh ds. \tag{11}$$

In this new regime, attrition is mechanically reduced to zero. However, teachers have to forego the opportunity of rematching and improving their outcome in the future, which might affect how they initially sort. A static model would not be able to quantify the potential adverse selection resulting from this policy.

Using the estimates described above, I simulate the realized outcome under the status quo and under this counterfactual scenario. In both simulations, I predict U and V using the estimated parameters, solve for the fixed point problem (4) to derive continuation values, and construct preferences using a random draw of unobserved taste shocks from the Type-I extreme value distribution.

Table 4 shows that this policy would have severe adverse sorting effects. As this policy forces teachers to commit and prevents them from improving their match in the future, this reinforces their distaste for schools located in remote areas. As a result, the urban-rural gap in teacher value-added is predicted to increase from 0.12σ to 0.15σ. This finding highlights a key trade-off between recruitment and retention. In the presence of a job ladder, retention policies that make rematching more difficult imply a significant decrease in the continuation value of accepting a job with undesirable attributes and generate adverse selection. To avoid the latter, such policies should compensate workers for preventing them to climb the ladder and improve their matching outcomes.

Table 4: Minimum Contract Length

	Data	Simulated	Counterfactual
	(1)	(2)	(3)
Panel A: Urban-Rural Inequality			
$\hat{\alpha}_i^*$ Urban	0.090	0.101	0.115
$\hat{\alpha}_i^*$ Rural	0.008	-0.018	-0.031
Panel B: Matched Characteristics			
Wage	1.688	1.681	1.687
Distance to Home	69.850	73.436	46.184
Amenity Index	-0.221	-0.209	-0.256
Travel Time	2.454	2.454	2.537
log(Pop)	7.803	7.937	7.714

NOTES. This table presents summary statistics about the distribution of matched characteristics in the observed data (column 1), in the simulated status quo match (column 2) and in the counterfactual scenario described in Section 6.3 (column 3). Panel A displays estimated shrunken average teacher effectiveness (ATE) by rurality. Panel B contains additional average matched school characteristics.

To quantify the magnitude of the compensation increase required to negate this adverse selection, I compute the difference between the utility that teachers get from their status quo match and their counterfactual match according to equation (11). This difference corresponds to the cost that each teacher would face by sticking to their status quo match instead of behaving optimally in the counterfactual scenario. It also indirectly provides an estimate of the value of being able to rematch and climb up the ladder. I find that the total cost of this policy would amount to a wage increase of S/282 monthly per capita (19% wage increase) while translating in a decrease of the urban-rural teacher value-added gap of 0.02σ. This suggests that incentivizing retention is costly when the opportunities to rematch are high, which is especially true for early career high value-added teachers.

7 Conclusion

This paper studies identification and estimation of an empirical model of dynamic two-sided matching with non-transferable utility. Matches are assumed to be stable in each period with respect to agents' lifetime utility while agents have rational expectations about their future match payoffs. Multiplicity of stable matches and of beliefs about the realization of future matches makes the problem highly intractable. By considering a large market approximation, I circumvent these issues and show existence and uniqueness of the limiting

distribution of match characteristics, which can be expressed analytically as a function of agents' payoff functions. This mapping is invertible such that the joint surplus can be identified and estimated from data on realized matches. Additional data on opportunity sets of one side of the market are sufficient to separately identify individual payoffs from the surplus.

I apply these tools to study the allocation of teachers in Peru and how dynamic contracts can curb the outflow of high value-added teachers from rural to urban areas. I find that mandating minimum contract lengths in rural areas to reduce attrition would create adverse selection and worsen urban-rural disparities in teacher value-added. Negating adverse selection would require a substantial increase in teacher compensation, suggesting that incentivizing retention might be too costly when the opportunities to rematch are high, which is especially true for early career high value-added teachers.

The tools developed in this paper are suited to matching markets where agents have limited ways of transferring utility or when transfers (i.e. wages) are fixed ex-ante and directly observed by the econometrician. This is particularly relevant for labor markets for public sector employees where wages are collectively bargained and are often rigid. Incorporating equilibrium wage setting or imperfectly transferable utility in the framework developed in this paper would be a fruitful avenue for future research.

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Supplementary Online Appendix

A Proofs

A.1 Proof of Proposition 1

Consider a match μ and suppose that there exists a blocking pair (i,j) such that $U_{ijt} > U_{i\mu_w(i,t)t}$ and $V_{ijt} > V_{\mu_m(j,t)jt}$. This would mean that $j \in M_{it}(\mu)$ and $U_{ijt} > U_{i\mu_w(i,t)t}$ which contradicts that $U_{i\mu_w(i,t)t} = \max_{k\in M_{it}(\mu)\cup\{0\}} U_{ikt}$. Now, suppose that individual rationality fails such that $U_{i0t} > U_{i\mu_w(i,t)t}$ or $V_{0jt} > V_{\mu_m(j,t)jt}$. In both cases, this would contradict that $U_{i\mu_w(i,t)t} = \max_{k\in M_{it}(\mu)\cup\{0\}} U_{ik}$ or $V_{\mu_m(j,t)jt} = \max_{l\in W_{jt}(\mu)\cup\{0\}} V_{ljt}$.

Now, suppose that for a given i, $U_{i\mu_w(i,t)t} < \max_{k\in M_{it}(\mu)\cup\{0\}} U_{ikt}$. This implies that there exists a school $k' \in M_{it}(\mu)\cup\{0\}$ such that $U_{ik't} > U_{i\mu_w(i,t)t}$. If $k' = 0$ this immediately contradicts stability. If $k' \in M_{it}(\mu)$ this implies that $V_{ik't} \geq V_{\mu_m(k',t)k't}$ and $U_{ik't} > U_{i\mu_w(i,t)t}$. If $V_{ik't} = V_{\mu_m(k',t)k't}$ this implies that $k' = \mu_w(i,t)$ and we reach a contradiction. Otherwise we have that $U_{ik't} > U_{i\mu_w(i,t)t}$ and $V_{ik't} > V_{\mu_m(k',t)k't}$ which contradicts stability. The argument is symmetric for the firm side. This proves part (i) of Proposition 1. Part (ii) is a direct implication of part (i) and Assumption 1.

A.2 Proof of Proposition 2

As $\bar{U}_{ijt}$ are considered fixed for all $j = 0,1,...,J$, I rewrite $U_{ijt} = u_{ijt} + \sigma\eta_{ijt}$ WLOG for simplicity. The proof of part (i) of Proposition 2 follows closely the proof of Lemma 3.1 in Menzel (2015).

$$\begin{aligned}
&\mathbb{P}(U_{ijt} \geq \max_{k=0,1,...,J} U_{ikt} | (u_{ikt})_{k=1}^J) \\
&\qquad = \int \mathbb{P}(U_{ijt} \geq U_{ikt}, k \in \mathcal{I} - \{j\} | (u_{ikt})_{k=1}^J, \eta_{ijt} = s) g(s) ds \\
&\qquad = \int \prod_{k\in\mathcal{I}-\{j\}} G(\sigma^{-1}(u_{ijt} - u_{ikt}) + s) g(s) ds \\
&\qquad = \int \prod_{k=1}^{2J} G(\sigma^{-1}(u_{ijt} - u_{ikt}) + s) \frac{g(s)}{G(s)} ds.
\end{aligned}$$

I apply the change of variables $s = a_J h + b_J$ where $a_J = a(b_J)$ and $b_J = G^{-1}(1 - J^{-1/2})$ and multiply by J on both sides:

$$\begin{aligned} &J\mathbb{P}(U_{ijt} \geq \max_{k=0,1,\ldots,J} U_{ikt} | (u_{ikt})_{k=1}^{J}) \\ &\quad = \int \exp\left(\frac{1}{J}\sum_{k=1}^{2J} J \log G(a_J(u_{ijt} - u_{ikt} + h) + b_J)\right) \frac{J a_J g(a_J h + b_J)}{G(a_J h + b_J)} dh. \end{aligned}$$

Following Resnick (1987) and under Assumption 1 we can show that:

$$J \log G(a_J(u_{ijt} - u_{ilt} + h) + b_J) \to -e^{-(u_{ijt} - u_{ikt} + h)},$$

$$\frac{J a_J g(a_J h + b_J)}{G(a_J h + b_J)} \to e^{-h}.$$

We thus have under Assumption 1

$$\begin{aligned} J\mathbb{P}(U_{ijt} \geq \max_{k=0,1,\ldots,J} U_{ikt} | (u_{ikt})_{k=1}^{J}) &= \int \exp\left(-\frac{1}{J}\sum_{k=1}^{2J} e^{-(u_{ijt} - u_{ikt} + h)}\right) e^{-h} dh + o(1) \\ &= \int \exp\left(-\frac{1}{J}\sum_{k=1}^{2J} e^{-h} e^{(u_{ikt} - u_{ijt})}\right) e^{-h} e^{-h} dh + o(1). \end{aligned}$$

I then apply a final change of variable $s = e^{-h}$ such that we get

$$\begin{aligned} J\mathbb{P}(U_{ijt} \geq \max_{k=0,1,\ldots,J} U_{ikt} | (u_{ikt})_{k=1}^{J}) &= \int_0^{+\infty} \exp\left(-\frac{1}{J}\sum_{k=1}^{2J} s e^{(u_{ikt} - u_{ijt})}\right) s ds + o(1) \\ &= \frac{\exp(u_{ijt})}{\frac{1}{J}\sum_{k=1}^{2J} \exp(u_{ikt})} + o(1). \end{aligned}$$

From this, we can finally show that

$$\begin{aligned} &J\mathbb{P}(U_{ijt} \geq \max_{k=0,\ldots,J} U_{ikt} | x_{it}, (z_{kt})_{k=1}^{J}, (\bar{U}_{ikt})_{k=0}^{J}) \\ &\qquad = \frac{\exp\left\{U_t(x_{it}, z_{jt}) + \beta \bar{U}_{ijt}\right\}}{\exp\left\{\beta \bar{U}_{i0t}\right\} + \frac{1}{J}\sum_{k=1}^{J} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\}} + o(1). \end{aligned}$$

Following the same steps, we can show that

$$\mathbb{P}(U_{i0t} \geq \max_{k=0,...,J} U_{ikt} | x_{it}, (z_{kt})_{k=1}^{J}, (\bar{U}_{ikt})_{k=0}^{J}) = \frac{\exp\left\{\beta \bar{U}_{i0t}\right\}}{\exp\left\{\beta \bar{U}_{i0t}\right\} + \frac{1}{J}\sum_{k=1}^{J} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\}} + o(1).$$

Using similar steps as in McFadden (1974), we can finally show that under Assumption 1:

$$\mathbb{E}\left[\max_{k=0,...,J} U_{ikt} | x_{it}, (z_{kt})_{k=1}^{J}, (\bar{U}_{ikt})_{k=0}^{J}\right] = \log\left(\exp\left\{\beta \bar{U}_{i0t}\right\} + \frac{1}{J}\sum_{k=1}^{J} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\}\right) + \log J + \gamma + o(1),$$

where γ is Euler's constant. This concludes the proof of Proposition 2.

A.3 Proof of Theorem 1 (i)

Throughout the rest of the proof, I set WLOG $\gamma_w = \gamma_m = 0$ for simplicity. I first show that the solutions to the fixed point problem described in equation (4) belong to a Banach space of continuous functions. Assume that there exists a set of $2 \times T$ functions Γ^*_{wt} and Γ^*_{mt} for all $t = 1, ..., T$ that solve the fixed point problem. I start by showing that these solutions are bounded from above. By definition of Ψ_{wt} and $\bar{U}^*_t$ and using that $\Gamma^*_{mt} \geq 0$ for all t, we can proceed by backward induction and show:

$$\begin{aligned}\Gamma^*_{wT}(x) = \Psi_{wT}[\Gamma^*_w, \Gamma^*_m](x) &= \int \frac{\exp\{U_T(x,s) + V_T(x,s)\}}{1 + \Gamma^*_{mT}(s)} m_T(s) ds \\ &\leq \int \exp\{U_T(x,s) + V_T(x,s)\} m_T(s) ds \\ &\leq \exp\{U^+ + V^+\},\end{aligned}$$

where U^+ and V^+ are the upper bounds in absolute value of the functions U_t and V_t for all t, respectively. From there, we can bound $\bar{U}^*_{T-1}$ as follows:

$$\begin{aligned}\bar{U}^*_{T-1}(x,z) &= \int \log\left(1 + \Gamma^*_{wT}(s)\right) w(s|x,z) ds \\ &\leq \log\left(1 + \exp\{U^+ + V^+\}\right).\end{aligned}$$

It is also immediate to show that $\bar{U}^*_{T-1} \geq 0$. Similar bounds can be derived on the firm side. We can then iterate this procedure and bound Γ^*_{wT-1} and $\bar{U}^*_{T-2}$:

$$\begin{aligned}\Gamma^*_{wT-1}(x) &= \int \frac{\exp\{U_{T-1}(x,h) + V_{T-1}(x,h) + \beta \bar{U}^*_{T-1}(x,h) + \delta \bar{V}^*_{T-1}(x,h)\}}{\exp\left\{\delta \bar{V}^*_{T-1}(x_0,h)\right\} + \Gamma^*_{mT-1}(h)} m^*_{T-1}(h)dh \\ &\leq \exp\left\{U^+ + V^+ + (\beta+\delta)\log(1+\exp\{U^+ + V^+\})\right\}.\end{aligned}$$

$$\begin{aligned}\bar{U}^*_{T-2}(x,z) = & \int \log\left(\exp\left\{\beta \bar{U}^*_{T-1}(s,z_0)\right\} + \Gamma^*_{wT-1}(s)\right) w(s|x,z)ds \\ \leq & \log\Bigg(\exp\left\{\beta \log\left(1+\exp\{U^+ + V^+\}\right)\right\} \\ & + \exp\left\{U^+ + V^+ + (\beta+\delta)\log(1+\exp\{U^+ + V^+\})\right\}\Bigg).\end{aligned}$$

Boundedness of Γ^*_{wt} is thus implied by boundedness of $\bar{U}^*_t$ and $\bar{V}^*_t$ which are implied by boundedness of Γ^*_{wt+1}, Γ^*_{mt+1}, $\bar{U}^*_{t+1}$ and $\bar{V}^*_{t+1}$. By induction we can thus show that boundedness of Γ^*_{wT} and Γ^*_{mt} implies boundedness of Γ^*_{wt} for all $t = 1, ..., T$. The same argument applies to Γ^*_{mt}. Continuity of Γ^*_{wt} and Γ^*_{mt} follows from continuity of U_t and V_t and that the integrals are nonnegative. Differentiability of Γ^*_{wt} and Γ^*_{mt} also follows from differentiability of U_t and V_t (see Assumption 1). We can thus restrict the spaces in which Γ^*_{wt} and Γ^*_{mt} belong to a Banach space of nonnegative bounded continuous functions which I denote $\mathcal{C}$.

I now turn to the proof that the mapping

$$(\log \Gamma_w, \log \Gamma_m) \mapsto (\log \Psi_m[\Gamma_w, \Gamma_m], \log \Psi_w[\Gamma_w, \Gamma_m]),$$

is a contraction. Consider two sets of functions $\Gamma = (\Gamma_w, \Gamma_m)$ and $\tilde{\Gamma} = (\tilde{\Gamma}_w, \tilde{\Gamma}_m)$ belonging to $\mathcal{C}^{2T}$. I show that there always exists a constant $\lambda < 1$ such that:

$$\left|\left| \log \Psi_w[\Gamma] - \log \Psi_w[\tilde{\Gamma}] \right|\right|_\infty \leq \lambda \left|\left| \log \Gamma - \log \tilde{\Gamma} \right|\right|_\infty.$$

The mean value inequality for vector valued functions defined on Banach spaces implies that:

$$\begin{aligned}&\left|\left| \log \Psi_w[\Gamma](x) - \log \Psi_w[\tilde{\Gamma}](x) \right|\right|_\infty \leq \\ &\qquad \sup_{a\in[0,1]} \left|\left| D \log \Psi_w[a \log \Gamma + (1-a)\log \tilde{\Gamma}](x) \right|\right|_\infty \left|\left| \log \Gamma(x) - \log \tilde{\Gamma}(x) \right|\right|_\infty.\end{aligned}$$

where $D \log \Psi_w$ are the Gateaux derivatives of $\log \Psi_w$. The rest of the proof consists in showing that these derivatives are strictly bounded below 1.

Starting with $t = 1$, I rewrite $\log \Psi_{w1}$ such that:

$$\begin{aligned}
&\log\ \Psi_{w1}[\log \Gamma](x) \\
&\quad = \log \int \frac{\exp\left\{U_1(x,h) + V_1(x,h) + \beta \bar{U}_1[\log \Gamma](x,h) + \delta \bar{V}_1[\log \Gamma](x,h)\right\}}{\exp\{\delta \bar{V}_1[\log \Gamma](x_0,h)\} + \exp\{\log \Gamma^*_{m1}(h)\}} m_1(h)dh,
\end{aligned}$$

where $\bar{U}_1$ and $\bar{V}_1$ are defined as

$$\bar{U}_1[\log \Gamma](x,z) = \int \log\left(\exp\left\{\beta \bar{U}_2[\log \Gamma](s,z_0)\right\} + \exp\{\log \Gamma_{w2}(s)\}\right) w(s|x,z)ds$$

$$\bar{V}_1[\log \Gamma](x,z) = \int \log\left(\exp\left\{\delta \bar{V}_2[\log \Gamma](x_0,s)\right\} + \exp\{\log \Gamma_{m2}(s)\}\right) m(s|x,z)ds.$$

The Gateaux derivative of $\log \Psi_{w1}$ with respect to $\log \Gamma_{m1}$ can be bounded in absolute value as follows

$$\begin{aligned}
&\left| -\frac{1}{\Psi_{w1}[\Gamma](x)} \right. \\
&\quad \left. \times \int \frac{\Gamma_{m1}(h) \exp\left\{U_1(x,h) + V_1(x,h) + \beta \bar{U}_1[\Gamma](x,h) + \delta \bar{V}_1[\Gamma](x,h)\right\}}{(\exp\{\delta \bar{V}_1[\Gamma](x_0,h)\} + \Gamma_{m1}(h))^2} m_1(h)dh \right| \\
&\quad \leq \quad \frac{\lambda_1}{\Psi_{w1}[\Gamma](x)} \int \frac{\exp\left\{U_1(x,h) + V_1(x,h) + \beta \bar{U}_1[\Gamma](x,h) + \delta \bar{V}_1[\Gamma](x,h)\right\}}{\exp\{\delta \bar{V}_1[\Gamma](x_0,h)\} + \Gamma_{m1}(h)} m_1(h)dh \\
&\quad = \quad \lambda_1.
\end{aligned}$$

where λ_1 is an upper bound of the ratio

$$0 \leq \frac{\Gamma_{m1}(z)}{\exp\{\delta \bar{V}_1[\Gamma](x_0,z)\} + \Gamma_{m1}(z)} \leq \lambda_1 < 1, \tag{12}$$

for all $z \in \mathcal{Z}$. A similar bound can be computed for the Gateaux derivative of $\log \Psi_{m1}$ with respect to $\log \Gamma_{w1}$.

Using a similar argument, we can show that the Gateaux derivative of $\log \Psi_{w1}$ with respect to $\log \Gamma_{mt}$ for $t > 1$ is bounded in absolute value by the upper bound of the following

expression:

$$\left|\delta D_{mt}\bar{V}_1[\log\Gamma](x,h) - \delta D_{mt}\bar{V}_1[\log\Gamma](x_0,h)\frac{\exp\{\delta\bar{V}_1(x_0,h)\}}{\exp\{\delta\bar{V}_1(x_0,h)\}+\Gamma^*_{m1}(h)}\right|,$$

where I define $D_{mt}\bar{V}_1$ as the Gateaux derivative of $\bar{V}_1$ with respect to $\log\Gamma_{mt}$. From equation (12), we can show that for all $1 < t < T$,

$$0 \leq D_{mt}\bar{V}_{t-1}[\log\Gamma](x,z) \leq \lambda_1 < 1, \tag{13}$$

and proceed by induction to show that for all $1 < t < T$:

$$0 \leq D_{mt+1}\bar{V}_{t-1}[\log\Gamma](x,z) \leq \delta\lambda_1\lambda_2 < 1,$$

where λ_2 is an upper bound of the ratio

$$0 \leq \frac{\exp\{\delta\bar{V}_t(x_0,s)\}}{\exp\{\delta\bar{V}_t(x_0,s)\}+\Gamma^*_{mt}(s)} \leq \lambda_2 < 1.$$

We can iterate this procedure to show that for all $1 \leq t < t' < T$:

$$0 \leq D_{mt'}\bar{V}_t[\log\Gamma](x,z) \leq \delta^{t'-t-1}\lambda_1\lambda_2^{t'-t-1} < 1. \tag{14}$$

For $t' = T$ we can easily verify that:

$$0 \leq D_{mT}\bar{V}_t[\log\Gamma](x,z) \leq \delta^{T-t-1}\lambda_1^2\lambda_2^{T-t-2} < 1.$$

This implies that we can bound from above the derivative of $\log\Psi_{w1}$ with respect to $\log\Gamma_{mt}$ for all $1 < t \leq T$ by 1 in absolute value. A similar argument can be used to bound the Gateaux derivatives of $\log\Psi_{w1}$ with respect to $\log\Gamma_w$. This also holds symmetrically for Ψ_{m1}. This completes the argument that the Gateaux derivatives of $\log\Psi_{w1}$ and $\log\Psi_{m1}$ are strictly bounded below 1.

I now consider $\log\Psi_{wt}$ such that $1 < t < T$. I rewrite $\log\Psi_{wt}$ such that:

$$\log\ \Psi_{wt}[\log\Gamma](x)$$

$$= \log \int \frac{\exp\left\{U_t(x,h) + V_t(x,h) + \beta \bar{U}_t[\log \Gamma](x,h) + \delta \bar{V}_t[\log \Gamma](x,h)\right\}}{\exp\{\delta \bar{V}_t[\log \Gamma](x_0,h)\} + \exp\{\log \Gamma_{mt}(h)\}} m_t[\log \Gamma](h)dh,$$

where m_t is defined as:

$$m_t[\log \Gamma](z) = \int_{\mathcal{X}} \int_{\mathcal{Z}} m(z|s,h) f_{t-1}[\log \Gamma](s,h)dhds + \int_{\mathcal{Z}} m(z|x_0,h) f_{t-1}[\log \Gamma](*,h)dh.$$

and f_t has the following expression:

$$f_t(x,z) = \frac{\exp\left\{U_t(x,z) + V_t(x,z) + \beta \bar{U}_t(x,z) + \delta \bar{V}_t(x,z)\right\} w_t[\log \Gamma](x) m_t[\log \Gamma](z)}{\left(\exp\left\{\beta \bar{U}_t(x,z_0)\right\} + \exp\{\log \Gamma_{wt}(x)\}\right)\left(\exp\left\{\delta \bar{V}_t(x_0,z)\right\} + \exp\{\log \Gamma_{mt}(z)\}\right)},$$

$$f_t(*,z) = \frac{\exp\left\{\delta \bar{V}_t(x_0,z)\right\}}{\exp\left\{\delta \bar{V}_t(x_0,z)\right\} + \exp\{\log \Gamma_{mt}(z)\}} m_t[\log \Gamma](z).$$

I consider the derivative of $\log \Psi_{wt}$ with respect to $\log \Gamma_{mt-1}$ and write is as:

$$-\frac{1}{\Psi_{wt}[\Gamma](x)} \int \frac{\exp\left\{U_t(x,h) + V_t(x,h) + \beta \bar{U}_t[\Gamma](x,h) + \delta \bar{V}_t[\Gamma](x,h)\right\}}{\exp\{\delta \bar{V}_t[\Gamma](x_0,h)\} + \Gamma_{mt}(h)} \times \frac{D_{mt-1} m_t[\Gamma](h)}{m_t[\Gamma](h)} m_t[\Gamma](h)dh,$$

where I define $D_{mt-1} m_t$ as the derivative of m_t with respect to $\log \Gamma_{mt-1}$. It follows directly from equations (13) and (14) that:

$$\left| \frac{D_{mt-1} m_t[\Gamma](h)}{m_t(h)} \right| < 1,$$

and that the derivative of $\log \Psi_{wt}$ with respect to $\log \Gamma_{mt-1}$ is strictly bounded from above in absolute value by 1. Similar steps can be used to show the same result for the Gateaux derivative of $\log \Psi_{wt}$ with respect to any $\log \Gamma_{mt'}$ or $\log \Gamma_{wt'}$ with $t \neq t'$. Symmetrical results apply for $\log \Psi_{mt}$. Overall, this implies that:

$$\sup_{a \in [0,1]} \left|\left| D \log \Psi_w[a \log \Gamma + (1-a) \log \tilde{\Gamma}](x) \right|\right|_\infty < 1,$$

which completes the proof that there exists a constant $\lambda < 1$ such that:

$$\left|\left| \log \Psi_w[\Gamma] - \log \Psi_w[\tilde{\Gamma}] \right|\right|_\infty \leq \lambda \left|\left| \log \Gamma - \log \tilde{\Gamma} \right|\right|_\infty,$$

where similar bounds can be established for Ψ_m.

I thus conclude that the mapping $(\log \Gamma_w, \log \Gamma_m) \mapsto (\log \Psi_w[\Gamma_w, \Gamma_m], \log \Psi_m[\Gamma_w, \Gamma_m])$ is a contraction. The second part of the statement of Theorem 1 part (i) is a direct implication of the Banach fixed point theorem.

A.4 Proof of Theorem 1 (ii)

A.4.1 Auxiliary Lemmas

Rate of Size of Feasible Choice Sets: Define, for $(\mu^*, \bar{U}^*, \bar{V}^*)$ satisfying Assumption 3, the number of firms feasible to worker i and the number of workers feasible to firm j in period t as

$$J^*_{wit} = \sum_{j=1}^{n_m} \mathbb{1}\{V_{ijt} \geq \max_{l \in W^*_{jt} \cup \{0\}} V_{ljt}\} \quad \text{and} \quad J^*_{mjt} = \sum_{i=1}^{n_w} \mathbb{1}\{U_{ijt} \geq \max_{k \in M^*_{it} \cup \{0\}} U_{ikt}\}.$$

Similarly, define the number of firms that worker i would accept and the number of workers that firm j would accept as

$$L^*_{wit} = \sum_{j=1}^{n_m} \mathbb{1}\{U_{ijt} \geq \max_{k \in M^*_{it} \cup \{0\}} U_{ikt}\} \quad \text{and} \quad L^*_{mjt} = \sum_{i=1}^{n_w} \mathbb{1}\{V_{ijt} \geq \max_{l \in W^*_{jt} \cup \{0\}} V_{ljt}\}.$$

I denote the upper bounds in absolute value of U_t and V_t as U^+ and V^+ and establish the following result:

Lemma 1 *Under Assumptions 1-3, there exists a set of constants J^+_{wt}, J^-_{wt}, J^+_{mt}, J^-_{mt}, L^+_{wt}, L^-_{wt}, L^+_{mt}, L^-_{mt} such that:*

$$n^{1/2} J^-_{wt} \leq J^*_{wit} \leq n^{1/2} J^+_{wt},$$

$$n^{1/2} J^-_{mt} \leq J^*_{mjt} \leq n^{1/2} J^+_{mt},$$

$$n^{1/2} L^-_{wt} \leq L^*_{wit} \leq n^{1/2} L^+_{wt},$$

$$n^{1/2} L^-_{mt} \leq L^*_{mjt} \leq n^{1/2} L^+_{mt},$$

for each $i = 1, ..., n_w$, $j = 1, ..., n_m$ and $t = 1, ..., T$ with probability approaching 1 as $n \to \infty$.

PROOF: As in Menzel (2015), we can define exogenous sets $\bar{W}_{jt} = \{i : U_{ijt} \geq U_{i0t}\}$ and $\bar{M}_{it} = \{j : V_{ijt} \geq V_{0jt}\}$ such that $W^*_{jt} \subset \bar{W}_{jt}$ and $M^*_{it} \subset \bar{M}_{it}$ as well as $W^\circ_{jt} = \{i : U_{ijt} \geq \max_{k \in \bar{M}_{it} \cup \{0\}} U_{ikt}\}$ and $M^\circ_{it} = \{j : V_{ijt} \geq \max_{l \in \bar{W}_{jt} \cup \{0\}} V_{ljt}\}$ such that $W^\circ_{jt} \subset W^*_{jt}$ and $M^\circ_{it} \subset M^*_{it}$.

From the definition of $\bar{M}_{it}$ and M°_{it}, we can construct the following bounds on J^*_{wit}:

$$J^\circ_{wit} = \sum_{j=1}^{n_m} \mathbb{1}\{j \in M^\circ_{it}\} \leq \sum_{j=1}^{n_m} \mathbb{1}\{j \in M^*_{it}\} \leq \sum_{j=1}^{n_m} \mathbb{1}\{j \in \bar{M}_{it}\} = \bar{J}_{wit}.$$

From there, using Proposition 2, we can start by period T and show that:

$$\begin{aligned}\mathbb{E}[\bar{J}_{wiT} | x_{iT}, (z_{kT})_{k=1}^{n_m}] &= \frac{1}{J} \sum_{j=1}^{n_m} \frac{\exp\{V_T(x_{iT}, z_{jT})\}}{1 + \frac{1}{J}\exp\{V_T(x_{iT}, z_{jT})\}} + o(1) \\ &\leq \frac{n_m}{J} \exp\{V^+\} + o(1),\end{aligned}$$

which implies under Assumption 2 that:

$$\mathbb{E}[\bar{J}_{wiT}] \leq n^{1/2} \exp\{V^+ + \gamma_m\} + o(1).$$

Proposition 2 and Assumption 2 also imply that the variance of $\bar{J}_{wiT}$ converges to zero which implies that:

$$n^{-1/2}(\bar{J}_{wiT} - \mathbb{E}[\bar{J}_{wiT}]) \to 0,$$

such that $J^*_{wiT} \leq n^{1/2} \exp\{V^+ + \gamma_m\}$ with probability approaching 1 as $n \to \infty$. Following the same steps, we can show symmetrically that:

$$J^*_{mjT} \leq n^{1/2} \exp\{U^+ + \gamma_w\},$$

$$L^*_{wiT} \leq n^{1/2} \exp\{U^+ + \gamma_m\},$$

$$L^*_{mjT} \leq n^{1/2} \exp\{V^+ + \gamma_w\},$$

with probability approaching 1 as $n \to \infty$. We now consider the lower bound J°_{wiT}. We can

again use Proposition 2 to show that:

$$\mathbb{E}[J^{\circ}_{wiT}|(x_{lT})_{l\in\bar{W}_{jT}},(z_{kT})_{k=1}^{n_m}] = \frac{1}{J}\sum_{j=1}^{n_m}\frac{\exp\{V_T(x_{iT},z_{jT})\}}{1+\frac{1}{J}\sum_{l\in\bar{W}_{jT}}\exp\{V_T(x_{lT},z_{jT})\}} + o(1)$$
$$\geq \frac{n_m}{J}\frac{\exp\{-V^+\}}{1+\frac{\bar{J}_{mjT}}{J}\exp\{V^+\}} + o(1).$$

Using the higher bound for J^*_{mjT} derived just above and Jensen's inequality, we can finally show that:

$$\mathbb{E}[J^{\circ}_{wiT}] \geq n^{1/2}\frac{\exp\{-V^+ + \gamma_m\}}{1+\exp\{V^+ + U^+ + \gamma_w\}} + o(1)$$

Proposition 2 and Assumption 2 also imply that the variance of $\bar{J}_{wit}$ converges to zero which implies that:

$$n^{-1/2}(J^{\circ}_{wiT} - \mathbb{E}[J^{\circ}_{wiT}]) \to 0,$$

such that $J^*_{wiT} \geq n^{1/2}\frac{\exp\{-V^+ + \gamma_m\}}{1+\exp\{V^+ + U^+ + \gamma_w\}}$ with probability approaching 1 as $n \to \infty$. Following the same steps, we can show that symmetrically, we have:

$$J^*_{mjT} \geq n^{1/2}\frac{\exp\{-U^+ + \gamma_w\}}{1+\exp\{V^+ + U^+ + \gamma_m\}},$$

$$L^*_{wiT} \geq n^{1/2}\frac{\exp\{-U^+ + \gamma_m\}}{1+\exp\{V^+ + U^+ + \gamma_m\}},$$

$$L^*_{mjT} \geq n^{1/2}\frac{\exp\{-V^+ + \gamma_w\}}{1+\exp\{V^+ + U^+ + \gamma_w\}},$$

with probability approaching 1 as $n \to \infty$. Having established that opportunity sets in period T grow at a rate $n^{1/2}$, Proposition 2 implies that, under Assumption 2, the expected utility of the chosen alternative in period T grows at a rate $\log n^{1/2}$, meaning that there exist constants $\bar{U}^-$, $\bar{U}^+$, $\bar{V}^-$ and $\bar{V}^+$ such that $\log n^{1/2}\bar{U}^- \leq \bar{U}^*_{ijt} \leq \log n^{1/2}\bar{U}^+$ and $\log n^{1/2}\bar{V}^- \leq \bar{V}^*_{ijt} \leq \log n^{1/2}\bar{V}^+$ for all $i = 0, ..., n_w$, $j = 0, ..., n_m$ and $t < T$. From there, using Proposition 2, we can show that for any $t < T$:

$$\mathbb{E}[\bar{J}_{wit}|x_{it},(z_{kt})_{k=1}^{n_m},(\bar{V}_{ikt})_{k=1}^{n_m},(\bar{V}_{0kt})_{k=1}^{n_m}]$$
$$= \frac{1}{J}\sum_{j=1}^{n_m}\frac{\exp\{V_t(x_{it},z_{jt}) + \delta\bar{V}_{ijt}\}}{\exp\{\delta\bar{V}_{0jt}\} + \frac{1}{J}\exp\{V_t(x_{it},z_{jt}) + \delta\bar{V}_{ijt}\}} + o(1),$$

which implies under Assumption 2 that:

$$\mathbb{E}[\bar{J}_{wit}] \leq n^{1/2} \exp\{V^+ + \delta\bar{V}^+ - \delta\bar{V}^- + \gamma_m\} + o(1).$$

Proposition 2 implies that the variance of $\bar{J}_{wit}$ converges to zero which implies that:

$$n^{-1/2}(\bar{J}_{wit} - \mathbb{E}[\bar{J}_{wit}]) \to 0.$$

We have thus established that $J^*_{wit} \leq n^{1/2} \exp\{V^+ + \delta\bar{V}^+ - \delta\bar{V}^- + \gamma_m\}$ with probability approaching 1 as $n \to \infty$. Following the same steps, we can show symmetrically that:

$$J^*_{mjt} \leq n^{1/2} \exp\{U^+ + \beta\bar{U}^+ - \beta\bar{U}^- + \gamma_w\},$$

$$L^*_{wit} \leq n^{1/2} \exp\{U^+ + \beta\bar{U}^+ - \beta\bar{U}^- + \gamma_m\},$$

$$L^*_{mjt} \leq n^{1/2} \exp\{V^+ + \delta\bar{V}^+ - \delta\bar{V}^- + \gamma_w\},$$

with probability approaching 1 as $n \to \infty$. Consider the lower bound J°_{wit}, we can again use Proposition 2 to show that:

$$\mathbb{E}[J^\circ_{wit} | (x_{lt})_{l \in \bar{W}_{jt}}, (z_{kt})_{k=1}^{n_m}, (\bar{V}_{ikt})_{k=1}^{n_m}, (\bar{V}_{0kt})_{k=1}^{n_m}]$$
$$= \frac{1}{J} \sum_{j=1}^{n_m} \frac{\exp\{V_t(x_{it}, z_{jt}) + \delta\bar{V}_{ijt}\}}{\exp\{\delta\bar{V}_{0jt}\} + \frac{1}{J}\sum_{l \in \bar{W}_{jt}} \exp\{V_t(x_{lt}, z_{jt}) + \delta\bar{V}_{ijt}\}} + o(1).$$

Using the higher bound for J^*_{mjt} derived just above and Jensen's inequality, we can finally show that:

$$\mathbb{E}[J^\circ_{wit}] \geq \frac{n_m}{J} \frac{\exp\{-V^+ + \delta\bar{V}^-\}}{\exp\{\delta\bar{V}^+\} + \frac{\bar{J}_{mjt}}{J} \exp\{V^+ + \delta\bar{V}^+\}} + o(1)$$
$$\geq n^{1/2} \frac{\exp\{-V^+ + \delta\bar{V}^- - \delta\bar{V}^+ + \gamma_m\}}{1 + \exp\{V^+ + U^+ + \delta\bar{U}^+ + \gamma_w\}} + o(1).$$

Proposition 2 implies again that the variance of J°_{wit} converges to zero which implies that:

$$n^{-1/2}(J^\circ_{wit} - \mathbb{E}[J^\circ_{wit}]) \to 0.$$

This establishes that $J^*_{wit} \geq n^{1/2} \frac{\exp\{-V^+ + \delta\bar{V}^- - \delta\bar{V}^+ + \gamma_m\}}{1 + \exp\{V^+ + U^+ + \beta\bar{U}^+ + \gamma_w\}}$ with probability approaching 1 as

$n \to \infty$. Following the same steps, we can show that symmetrically, we have:

$$J^*_{mjt} \geq n^{1/2} \frac{\exp\{-U^+ + \beta \bar{U}^- - \beta \bar{U}^+ + \gamma_w\}}{1 + \exp\{V^+ + \delta \bar{V}^+ + U^+ + \gamma_m\}},$$

$$L^*_{wit} \geq n^{1/2} \frac{\exp\{-U^+ + \beta \bar{U}^- - \beta \bar{U}^+ + \gamma_m\}}{1 + \exp\{V^+ + \delta \bar{V}^+ + U^+ + \gamma_m\}},$$

$$L^*_{mjt} \geq n^{1/2} \frac{\exp\{-V^+ + \delta \bar{V}^- - \delta \bar{V}^+ + \gamma_w\}}{1 + \exp\{V^+ + U^+ + \beta \bar{U}^+ + \gamma_w\}},$$

with probability approaching 1 as $n \to \infty$. This concludes the proof of Lemma 1.

Exogeneity of Feasible Choice Sets: We now need to show that as $n \to \infty$, the dependence between agents taste shocks and opportunity sets vanishes. As taste shocks are independent across periods under Assumption 1, the dependence between unobserved preferences and opportunity sets can only arise within period. The proof thus mirrors very closely Menzel (2015), which proves the same result in the static case.

For the rest of the proof, I define the following set of indicator functions $E^*_{ijt} = \mathbb{1}\{i \in W^*_{jt}\}$ and $D^*_{ijt} = \mathbb{1}\{j \in M^*_{it}\}$ for all workers $i = 1, ..., n_w$ and firms $j = 1, ..., n_m$. The first result to establish is that the probability that changing one availability indicator affects another agents' opportunity set converges to zero as $n \to \infty$. I first prove the following result:

Lemma 2 *Suppose Assumption 1 and 2 hold and suppose we change one availability indicator E^*_{ijt} exogenously to $\tilde{E}_{ijt} = 1 - E^*_{ijt}$ and then iterate the deferred acceptance algorithm from this point until convergence. Denote the resulting availability indicators $\{\tilde{E}_{lkt}, \tilde{D}_{lkt} : l = 1, ..., n_w, k = 1, ..., n_m, t = 1, ..., T\}$. We have for any worker l, firm k and period t:*

(i). $\mathbb{P}(\tilde{D}_{lt} \neq D^*_{lt} | D^*_{lt}, D^*_{ijt} = 0) = \mathbb{P}(\tilde{E}_{kt} \neq E^*_{kt} | E^*_{lt}, D^*_{ijt} = 0) = 0$

(ii). There exist constants $\bar{a} < \infty$ and $0 < \lambda < 1$ such that:

$$\mathbb{P}(\tilde{D}_{lt} \neq D^*_{lt} | D^*_{lt}, D^*_{ijt} = 1) \leq n^{-1/2} \frac{\bar{a}}{1 - \lambda},$$

$$\mathbb{P}(\tilde{E}_{kt} \neq E^*_{kt} | E^*_{lt}, E^*_{ijt} = 1) \leq n^{-1/2} \frac{\bar{a}}{1 - \lambda}.$$

The same result holds for an exogenous change of D_{ijt} to $\tilde{D}_{ijt} = 1 - D_{ijt}$.

PROOF: Fix a period t. Suppose we change E^*_{jit} exogenously to $\tilde{E}_{jit} = 1 - E_{jit}$ and that we iterate the deferred acceptance algorithm from this stage. This will only trigger a chain of rematches if this affects the indirect utility of either i or j. Suppose $D^*_{ijt} = 0$ and that $E^*_{ijt} = 0$ meaning that firm j is not feasible to worker i and vice versa. Suppose now that $\tilde{E}_{jit} = 1 - E^*_{ijt} = 1$, meaning that suddenly worker i's preference for firm j increase such that worker i becomes feasible for firm j. This will not affect the indirect utility of firm j nor worker i given that firm j is not feasible to worker i. This change will thus not trigger a chain of rematches. A similar argument can be used in the case where E^*_{ijt} changes from 1 to $\tilde{E}_{jit} = 1 - E^*_{ijt} = 0$. This establishes part (i) of Lemma 2.

Now suppose that $D^*_{ijt} = 1$ such that if $\tilde{E}_{ijt} = 1 - E^*_{ijt} = 1$, now firm j and worker i will want to rematch together or if $\tilde{E}_{ijt} = 1 - E^*_{ijt} = 0$ firm j and worker i will break their current match. This will trigger a chain of rematches than can potentially cycle back to worker i or firm j's opportunity set. I start by showing that, at each step s of these subsequent rematches, there is at most one indicator in the vector $D^{(s)}_{lt}$ corresponding to a firm k with $E^{(s)}_{lkt} = 1$ that will change. The idea of the proof is the following: suppose that a given worker l matched to firm k in step $(s-1)$ becomes unavailable to firm k in step s. This firm will then replace this worker by its most preferred feasible alternative, which will only change the availability indicator of this firm to this newly hired worker. On the other hand, if a given worker becomes available to a firm while this firm prefers this worker to its matched employee, then it will replace them with this new employee, making this firm unavailable to the laid off employee. In both cases, this will only change at most one availability indicator among the workers who are willing to match with this firm. Note that in each of these steps, there is a chance that the chain is terminated if the next preferred feasible option is the outside option. A similar argument can be used symmetrically from the workers' perspective.

The rest of the proof now consists in bounding the probability that the chain is terminated by either (i) firm k or worker l preferring the outside option to any other option in their opportunity set or (ii) a change in availability indicators of worker k D_{kt}. I define as μ^s the state of the match in iteration s of the deferred acceptance algorithm following an exogenous change of E_{ijt} to $\tilde{E}_{ijt} = 1 - E_{ijt}$. The first step bounds the probability that the chain is terminated by the outside option at stage s.

I start from the following observation: given that $W^\circ_{kt} \subset W^*_{kt} \subset \bar{W}_{kt}$, we have from

Proposition 2 and Lemma 1 that for any firm k and worker l:

$$\begin{aligned}
&\mathbb{P}(V_{lkt} > \max_{i \in W_{kt}(\mu^s) \cup \{0\}} V_{ikt} | x_{lt}, z_{kt}, \bar{V}_{lkt}) \\
&\qquad \geq \mathbb{P}(V_{lkt} > \max_{i \in \bar{W}_{kt} \cup \{0\}} V_{ikt} | x_{lt}, z_{kt}, \bar{V}_{lkt}) \\
&\qquad = n^{-1/2} \frac{\exp(V_t(x_{lt}, z_{kt}) + \delta \bar{V}_{lkt})}{\exp\{\delta \bar{V}_{0kt}\} + \frac{1}{J} \sum_{i \in \bar{W}_{kt}} \exp(V_t(x_{it}, z_{kt}) + \delta \bar{V}_{ikt})} + o(1) \\
&\qquad \geq n^{-1/2} \frac{\exp(V_t(z_{kt}, x_{lt}) + \delta \bar{V}_{lkt})}{\exp\{\delta \bar{V}^+\} + \exp(U^+ + V^+ + \delta \bar{V}^+ + \beta \bar{U}^+ + \gamma_w)} + o(1).
\end{aligned}$$

This implies that, conditional on D^*_{it} and as n approaches infinity:

$$\begin{aligned}
&\mathbb{P}(V_{0kt} > \max_{l \in W_{kt}(\mu^s) \cup \{0\}} V_{lkt} | D^*_{it}, x_{it}, z_{kt}) \\
&\geq \frac{1}{\exp\{\delta \bar{V}^+ - \delta \bar{V}^-\} + \exp(\bar{U} + \bar{V} + \beta \bar{V}^+ - \delta \bar{V}^- + \beta \bar{U}^+ + \gamma_w)} =: p_s.
\end{aligned}$$

Following now the same steps as Menzel (2015), we have, by Bayes law that:

$$\mathbb{P}(V_{0kt} > \max_{l \in W_{kt}(\mu^s) \cup \{0\}} V_{ljt} | D^*_{lt}, \tilde{D}^{(s)}_{lkt} = 1, x_{lt}, z_{kt}) \geq \frac{L^-_{mt} p_s}{L^+_{mt}(1 - p_s) + L^-_{mt} p_s} > 0,$$

where L^+_{mt} and L^-_{mt} are respectively the upper and lower bounds on L^*_{mjt} taken from Lemma 1. This means that the probability of choosing the outside option at any iteration s is strictly bounded away from below by 0 implying that the probability that the chain is not terminated at stage s is bounded away from 1.

Now we bound the probability that the chain leads to a change in D_{lt} at stage s. We can thus bound the following probability using Proposition 2 and Lemma 1:

$$\begin{aligned}
&\mathbb{P}(V_{lkt} > \max_{i \in W_{kt}(\mu^s) \cup \{0\}} V_{ikt} | x_{lt}, z_{kt}) \\
&\qquad \leq \mathbb{P}(V_{lkt} > \max_{i \in W^\circ_{kt} \cup \{0\}} V_{ikt} | x_{lt}, z_{kt}) \\
&\qquad = n^{-1/2} \frac{\exp(V(z_{kt}, x_{lt}) + \delta \bar{V}_{lkt})}{\exp\{\delta \bar{V}_{0kt}\} + \frac{1}{J} \sum_{i \in W^\circ_{kt}} \exp(V(z_{kt}, x_{it}) + \delta \bar{V}_{ikt})} + o(1) \\
&\qquad \leq n^{-1/2} \exp(V^+ + \delta \bar{V}^+ - \delta \bar{V}^-) + o(1).
\end{aligned}$$

This implies that for n sufficiently large, we have:

$$\mathbb{P}(\tilde{D}_{lt}^{(s)} \neq D_{lt}^{*} | D_{lt}^{*}, \tilde{D}_{lkt}^{(s)} = 1, x_{lt}, z_{kt}) \leq \frac{n^{-1/2}\exp(V^{+} + \delta\bar{V}^{+} - \delta\bar{V}^{-})L_{mt}^{+}}{n^{-1/2}\exp(V^{+} + \delta\bar{V}^{+} - \delta\bar{V}^{-})L_{mt}^{+} + L_{mt}^{-}}$$
$$\leq n^{-1/2}\exp(V^{+} + \delta\bar{V}^{+} - \delta\bar{V}^{-})\frac{L_{mt}^{+}}{L_{mt}^{-}} = n^{-1/2}\bar{a}.$$

Using the law of total probability, we can thus bound as $n \to \infty$ the conditional probability that $\tilde{D}_{lt} \neq D_{lt}^{*}$

$$\mathbb{P}(\tilde{D}_{lt} \neq D_{lt}^{*} | D_{lt}^{*}) \leq \sum_{s=1}^{\infty} \lambda^{s} n^{-1/2}\bar{a} \leq \frac{n^{-1/2}\bar{a}}{1-\lambda},$$

which proves part (ii) of Lemma 2.

From there, I state the main result that the dependence between taste shocks and agents' opportunity sets vanishes as $n \to \infty$. I first define the joint distribution of $\eta_{it} = (\eta_{i1t}, ..., \eta_{in_m t})'$, $\epsilon_{jt} = (\epsilon_{1jt}, ..., \epsilon_{n_w jt})'$ and the availability indicators D_{it}^{W}, E_{jt}^{W}, D_{it}^{M}, E_{jt}^{M} corresponding to the worker-optimal and the firm-optimal stable matches. Note that I consider these two specific matches since the worker-optimal and firm-optimal stable matches are defined with probability 1 conditional on the realization of the taste shocks η_{it} and ϵ_{jt}. Indeed, the distribution of availability indicators arising from an arbitrary stable match D_{it}^{*} would not be well defined. I also define: $D_{i,-j,t}^{W} = (D_{i1t}^{W}, ..., D_{i(j-1)t}^{W}, D_{i(j+1)t}^{W}, ..., D_{in_m t}^{W})$ and $E_{-i,j,t} = (E_{1jt}^{W}, ..., E_{(i-1)jt}^{W}, E_{(i+1)jt}^{W}, ..., E_{n_w jt}^{W})$ with analogous notations for the firm optimal match. I then define the conditional c.d.f.s for $\boldsymbol{d} \in \{0,1\}^{n_m}$ and $\boldsymbol{e} \in \{0,1\}^{n_w - 1}$:

$$G_{\eta|D}^{W}(\eta|\boldsymbol{d}) = \mathbb{P}(\eta_{it} \leq \eta | D_{it}^{W} = \boldsymbol{d}),$$

$$G_{\eta,\epsilon|D,E}^{W}(\eta, \epsilon|\boldsymbol{d}, \boldsymbol{e}) = \mathbb{P}(\eta_{it} \leq \eta, \epsilon_{jt} \leq \epsilon | D_{i,-j,t}^{W} = \boldsymbol{d}, E_{-i,j,t}^{W} = \boldsymbol{e}),$$

with analogous definitions for the firm-optimal stable match and associated p.d.f.s $g_{\eta|D}^{W}$ and $g_{\eta,\epsilon|D,E}^{W}$. The main result is the following:

Lemma 3 *Under Assumption 1 and 2, we have:*

(i). $g^W_{\eta|D}$ and $g^M_{\eta|D}$ satisfy:

$$\lim_n \Big| \frac{g^W_{\eta|D}(\eta|D^W_{it})}{g_\eta(\eta)} - 1 \Big| = \lim_n \Big| \frac{g^M_{\eta|D}(\eta|D^M_{it})}{g_\eta(\eta)} - 1 \Big| = 0.$$

(ii). $g^W_{\eta,\epsilon|D,E}$ and $g^M_{\eta,\epsilon|D,E}$ satisfy:

$$\lim_n \Big| \frac{g^W_{\eta|D}(\eta,\epsilon|D^W_{i,-j,t}, E^W_{-i,j,t})}{g_{\eta,\epsilon}(\eta,\epsilon)} - 1 \Big| = \lim_n \Big| \frac{g^M_{\eta|D}(\eta,\epsilon|D^M_{i,-j,t}, E^M_{-i,j,t})}{g_{\eta,\epsilon}(\eta,\epsilon)} - 1 \Big| = 0.$$

The same results holds for the firm side of the market.

PROOF: Let $g^W_{\eta,D}$ be the joint p.d.f. of taste shocks and availability indicators under the worker optimal stable match. We can rewrite, by definition of a conditional density:

$$\frac{g^W_{\eta|D}(\eta|D^W_{it})}{g_\eta(\eta)} = \frac{g^W_{\eta,D}(\eta, D^W_{it})}{g_\eta(\eta)P(D^W_{it})} = \frac{P(D^W_{it}|\eta_{it}=\eta)g_\eta(\eta)}{g_\eta(\eta)P(D^W_{it})} = \frac{P(D^W_{it}|\eta_{it}=\eta)}{P(D^W_{it})}.$$

I then follow similar steps as in Menzel (2015) to show that:

$$\Big| \frac{P(D^W_{it}|\eta_{it}=\eta)}{P(D^W_{it})} - 1 \Big| \leq \sup_{\eta_1,\eta_2} \Big| \frac{P(D^W_{it}|\eta_{it}=\eta_1)}{P(D^W_{it}|\eta_{it}=\eta_2)} - 1 \Big|,$$

such that I only need to bound the probability that shifting η_{it} from η_1 to η_2 changes worker i's opportunity set. We know from Lemma 2 that changing an availability indicator will trigger a chain of rematches that could change worker i's opportunity set with probability less than $\frac{n^{-1/2}\bar{a}}{1-\lambda}$ as n approaches infinity. Here, we can show that shifting agent i's taste shocks would trigger at most two chains of rematches. Indeed, if the shift in taste shocks makes agent i prefers firm l with $D_{ilt}=1$ instead of her current employer firm j, this changes both E_{ijt} from 1 to 0 and E_{ilt} from 0 to 1. Thus, this would trigger two chains of rematches where both firm j and the worker which was displaced from firm l by worker i would need to find a new match. We can thus conclude that:

$$\frac{P(D^W_{it}|\eta_{it}=\eta_1)}{P(D^W_{it}|\eta_{it}=\eta_2)} - 1 \leq 2\frac{n^{-1/2}\bar{a}}{1-\lambda},$$

as $n \to \infty$ which can be shown to hold also in absolute value. As the right hand side

converges to 0 as $n \to \infty$, this proves the first part of claim (i). The same result holds symetrically for the firm side.

For part (ii), note that the argument can be extended in a similar way. If you change both firm j and worker i's taste shocks this can trigger at most 4 chains of rematches such that we can bound the probability of a shift in opportunity sets by $n^{-1/2}\frac{4\bar{a}}{1-\lambda}$ which can be made arbitrarily close to 0 as n approaches infinity.

Approximation Inclusive Values: I define $I^M_{wit}[\bar{U},\bar{V}] = I_{wit}[M^M_{it}[\bar{U},\bar{V}],\bar{U}]$ and $I^M_{mjt}[\bar{U},\bar{V}] = I_{mjt}[W^M_{jt}[\bar{U},\bar{V}],\bar{V}]$ as the inclusive values that would arise from the firm-optimal stable match conditional on $\bar{U}$ and $\bar{V}$ in period t. Similarly, I define $I^W_{wit}[\bar{U},\bar{V}]$ and $I^W_{mjt}[\bar{U},\bar{V}]$ as the inclusive values that would arise from the worker-optimal stable match and $I^*_{wit}[\bar{U},\bar{V}]$ and $I^*_{mjt}[\bar{U},\bar{V}]$ as the inclusive values that would arise from any stable match.

I also define $\bar{U}^M[\bar{U},\bar{V}]$ and $\bar{V}^M[\bar{U},\bar{V}]$ as the continuation values that would arise under the expectation that the realized match in the next period will be firm-optimal conditional on subsequent continuation values $\bar{U}$ and $\bar{V}$. I define $\bar{U}^W$ and $\bar{V}^W$ analogously for the worker-optimal stable match and $\bar{U}^*$ and $\bar{V}^*$ for any stable match.

I define $\hat{\Gamma}^M_{wt}$ and $\hat{\Gamma}^M_{mt}$ as the firm-optimal *expected inclusive value functions* of workers and firms, respectively, such that

$$\hat{\Gamma}^M_{wt}[\bar{U},\bar{V}](x_{it}) = \frac{1}{n}\sum_{k=1}^{n_m} \frac{\exp\left\{U_t(x_{it},z_{kt}) + V_t(x_{it},z_{kt}) + \beta\bar{U}_{ikt} + \delta\bar{V}_{ikt}\right\}}{\exp\left\{\delta\bar{V}_{0kt}\right\} + I^M_{mkt}[\bar{U},\bar{V}]},$$

$$\hat{\Gamma}^M_{mt}[\bar{U},\bar{V}](z_{jt}) = \frac{1}{n}\sum_{l=1}^{n_w} \frac{\exp\left\{U_t(x_{lt},z_{jt}) + V_t(x_{lt},z_{jt}) + \beta\bar{U}_{ljt} + \delta\bar{V}_{ljt}\right\}}{\exp\left\{\beta\bar{U}_{l0t}\right\} + I^M_{wlt}[\bar{U},\bar{V}]}.$$

Lemma 4 *Under Assumption 1 and 2 and for an arbitrary vector of continuation values $(\bar{U},\bar{V})$ bounded in absolute value:*
(i). For all $i = 1,...,n_w$, $j = 1,...,n_m$ and $t = 1,...,T$:

$$I^M_{wit}[\bar{U},\bar{V}] \geq \hat{\Gamma}^M_{wt}[\bar{U},\bar{V}](x_{it}) + o_p(1) \quad \textit{and} \quad I^M_{mjt}[\bar{U},\bar{V}] \leq \hat{\Gamma}^M_{mt}[\bar{U},\bar{V}](z_{jt}) + o_p(1),$$

where the analogous result holds for the worker-optimal stable match with the side of inequalities reversed.
(ii). If the weight functions $\omega(x,z) \geq 0$ are bounded and form a Glivenko-Cantelli class in

x, then

$$\sup_{x\in\mathcal{X}} \frac{1}{n}\sum_{j=1}^{n_m} \omega(x, z_{jt})(I^M_{mjt}[\bar{U},\bar{V}] - \hat{\Gamma}^M_{mt}[\bar{U},\bar{V}](z_{jt})) \leq o_p(1),$$

and

$$\inf_{z\in\mathcal{Z}} \frac{1}{n}\sum_{i=1}^{n_w} \omega(x_{it}, z)(I^M_{wit}[\bar{U},\bar{V}] - \hat{\Gamma}^M_{wt}[\bar{U},\bar{V}](x_{it})) \geq o_p(1).$$

The analogous conclusion holds for the worker-optimal stable match where the sign of the inequalities is reversed and if $\omega(x,z) \geq 0$ *are bounded and form a Glivenko-Cantelli class in* z*.*

PROOF: I first define the conditional probability that worker i chooses firm j when facing opportunity set M^M arising from the firm-optimal stable match in period t:

$$\begin{aligned}&\Lambda^M_{wt}(x, z, M^M, \bar{U})\\ &\quad = \mathbb{P}(U_{ijt} \geq \max_{k\in M^M_{it}[\bar{U},\bar{V}]\cup\{0\}} U_{ikt} | M^M_{it}[\bar{U},\bar{V}] = M^M, x_{it} = x, z_{jt} = z, \bar{U}_{ijt} = \bar{U}).\end{aligned}$$

I also define the conditional choice probabilities that would arise fixing exogenously opportunity sets to M as:

$$\Lambda_{wt}(x, z, M, \bar{U}) = \mathbb{P}(U_{ijt} \geq \max_{k\in M\cup\{0\}} U_{ikt} | x_{it} = x, z_{jt} = z, \bar{U}_{ijt} = \bar{U}).$$

Lemma 3 implies that there exists a selection of stable matchings $\mu^*[\bar{U},\bar{V}]$ such that $M^*_{it}[\bar{U},\bar{V}] = M^M_{it}[\bar{U},\bar{V}]$ and taste shifters are independent of $M^*_{it}[\bar{U},\bar{V}]$ with probability approaching 1 such that:

$$J\Lambda^M_{wt}(x, z, M^M_{it}[\bar{U},\bar{V}], \bar{U}) \leq J\Lambda_{wt}(x, z, M^M_{it}[\bar{U},\bar{V}], \bar{U}) + o_p(1). \tag{15}$$

Using Proposition 2, we can then show that for $i = 1, ..., n_w$, $l_1 = 1, ..., n_m$ and $l_2 \neq l_1$:

$$\mathbb{E}\left[J(D^M_{il_1t} - \Lambda^M_{mt}(x_{it}, z_{l_1t}, I^M_{ml_1t}[\bar{U},\bar{V}], \bar{V}_{il_1t}))\middle| I^M_{ml_1t}[\bar{U},\bar{V}], x_{it}, z_{l_1t}, \bar{V}_{il_1t}\right] \to 0,$$

and

$$\begin{aligned}&\mathbb{E}\Big[J^2(D^M_{il_1t} - \Lambda^M_{mt}(x_{it}, z_{l_1t}, I^M_{ml_1t}[\bar{U},\bar{V}], \bar{V}_{il_1t}))(D^M_{il_2t}\\ &- \Lambda^M_{mt}(x_{it}, z_{l_2t}, I^M_{ml_2t}[\bar{U},\bar{V}], \bar{V}_{il_2t}))\Big| I^M_{ml_1t}[\bar{U},\bar{V}], I^M_{ml_2t}[\bar{U},\bar{V}], x_{it}, z_{l_1t}, z_{l_2t}, \bar{V}_{il_1t}, \bar{V}_{il_2t}\Big] \to 0.\end{aligned}$$

Therefore, since $\exp(U_t(x_{it}, z_{jt}) + \beta \bar{U}_{ijt})$ is bounded under Assumption 1, we can conclude that:

$$\text{Var}\left(\frac{1}{n}\sum_{k=1}^{n_m} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\} J(D^M_{ikt} - \Lambda^M_{mt}(x_{it}, z_{kt}, I^M_{mkt}[\bar{U}, \bar{V}], \bar{V}_{ikt})\right) \to 0,$$

which implies that:

$$\frac{1}{n}\sum_{k=1}^{n_m} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\} J(D^M_{ikt} - \Lambda^M_{mt}(x_{it}, z_{kt}, I^M_{mkt}[\bar{U}, \bar{V}], \bar{V}_{ikt}) = o_p(1).$$

Given that from Proposition 2 and equation (15):

$$J\Lambda^M_{mt}(x, z, I^M_{mjt}[\bar{U}, \bar{V}], \bar{V}_{ijt}) \geq \frac{\exp\left\{V_t(x, z) + \delta \bar{V}_{ijt}\right\}}{\exp\left\{\delta \bar{V}_{0jt}\right\} + I^M_{mjt}[\bar{U}, \bar{V}]} + o_p(1),$$

this implies that:

$$\begin{aligned}
&\frac{1}{n}\sum_{k=1}^{n_m} \exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt}\right\} J\left(D^M_{ikt} - \frac{\exp\left\{V_t(x_{it}, z_{kt}) + \delta \bar{V}_{ikt}\right\}}{\exp\left\{\delta \bar{V}_{0kt}\right\} + I^M_{mkt}[\bar{U}, \bar{V}]}\right) \\
&= I^M_{wit}[\bar{U}, \bar{V}] - \frac{1}{n}\sum_{k=1}^{n_m} \frac{\exp\left\{U_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt} + V_t(x_{it}, z_{kt}) + \delta \bar{V}_{ikt}\right\}}{\exp\left\{\delta \bar{V}_{0kt}\right\} + I^M_{mkt}[\bar{U}, \bar{V}]} \geq o_p(1).
\end{aligned}$$

This proves the first claim of part (i) of Lemma 4. Similar steps can be used to bound inclusive values on the firm side and for the worker optimal sequence of stable matches.

Part (ii) follows from part (i) of the Lemma and the boundedness condition on ω which implies pointwise convergence. The Glivenko-Cantelli condition on ω then implies uniform convergence. This concludes the proof of Lemma 4.

The next step consists in establishing uniform convergence with respect to $\Gamma_{wt} \in \mathcal{T}_{wt}$ and $\Gamma_{mt} \in \mathcal{T}_{mt}$ of the fixed point mappings $\hat{\Psi}_{wt}$ and $\hat{\Psi}_{mt}$ to their population counterparts. I define:

$$\hat{\Psi}_{wt}[\Gamma_w, \Gamma_m](x) = \frac{1}{n}\sum_{j=1}^{n_m} \psi_{wt}(z_{jt}, x; \Gamma_w, \Gamma_m),$$

where ψ_{wt} is defined as:

$$\psi_{wt}(z_{jt}, x; \Gamma_w, \Gamma_m) = \frac{\exp\left\{U_t(x, z_{jt}) + V_t(x, z_{jt}) + \beta \bar{U}_t[\Gamma_w](x, z_{jt}) + \delta \bar{V}_t[\Gamma_m](x, z_{jt})\right\}}{\exp\left\{\delta \bar{V}_t[\Gamma_m](x_{0t}, z_{jt})\right\} + \Gamma_{mt}(z_{jt})}.$$

Similarly, I define:

$$\hat{\Psi}_{mt}[\Gamma_w, \Gamma_m](z) = \frac{1}{n} \sum_{i=1}^{n_w} \psi_{mt}(z, x_{it}; \Gamma_w, \Gamma_m),$$

where ψ_{mt} is defined as:

$$\psi_{mt}(z, x_{it}; \Gamma_w, \Gamma_m) = \frac{\exp\left\{U_t(x_{it}, z) + V_t(x_{it}, z) + \beta \bar{U}_t[\Gamma_w](x_{it}, z) + \delta \bar{V}_t[\Gamma_m](x_{it}, z)\right\}}{\exp\left\{\beta \bar{U}_t[\Gamma_w](x_{it}, z_{0t})\right\} + \Gamma_{wt}(x_{it})}.$$

I define the class of functions $\mathcal{F}_w : \{\psi_w(., x; \Gamma_w, \Gamma_m) : x \in \mathcal{X}, (\Gamma_w, \Gamma_m) \in \mathcal{T}\}$ and $\mathcal{F}_m : \{\psi_m(z, ; \Gamma_w, \Gamma_m) : z \in \mathcal{Z}, (\Gamma_w, \Gamma_m) \in \mathcal{T}\}$.

Lemma 5 *Under Assumption 1:*
(i). The classes of functions $\mathcal{F}_w$ and $\mathcal{F}_w$ are Glivenko-Cantelli.
(ii). As $n \to \infty$ and for all t:

$$(\hat{\Psi}_{wt}[\Gamma_w, \Gamma_m](x), \hat{\Psi}_{mt}[\Gamma_w, \Gamma_m](z)) \to (\Psi_{wt}[\Gamma_w, \Gamma_m](x), \Psi_{mt}[\Gamma_w, \Gamma_m](z))$$

uniformly in $(\Gamma_w, \Gamma_m) \in \mathcal{T}$, and $(x, z) \in \mathcal{X} \times \mathcal{Z}$.

PROOF: Under Assumption 1, $\exp\{U(x, z) + V(x, z)\}$ is Lipschitz in x and z such that this class of functions is Glivenko-Cantelli. Γ_{mt} and Γ_{wt} are bounded and have bounded $p \geq 1$ derivatives for all t which makes the class of functions $\mathcal{T}$ Glivenko-Cantelli. Finally, m and w are continuous densities such that $\bar{U}_t$ and $\bar{V}_t$ are continuous and bounded. Theorem 3 in van der Vaart and Wellner (2000) thus implies claim (i) of Lemma 5. Part (ii) of Lemma 5 is a direct implication of part (i).

A.4.2 Proof of Theorem 1 (ii)

I first apply Lemma 4 to show that:

$$\hat{\Gamma}^M_{wt}[\bar{U}, \bar{V}](x_{it}) = \frac{1}{n} \sum_{k=1}^{n_m} \frac{\exp\left\{U_t(x_{it}, z_{kt}) + V_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt} + \delta \bar{V}_{ikt}\right\}}{\exp\left\{\delta \bar{V}_{0kt}\right\} + I^M_{mkt}[\bar{U}, \bar{V}]}$$

$$\geq \frac{1}{n}\sum_{k=1}^{n_m} \frac{\exp\left\{U_t(x_{it}, z_{kt}) + V_t(x_{it}, z_{kt}) + \beta \bar{U}_{ikt} + \delta \bar{V}_{ikt}\right\}}{\exp\left\{\delta \bar{V}_{0kt}\right\} + \hat{\Gamma}^M_{mt}[\bar{U}, \bar{V}](z_{kt})} + o_p(1).$$

Analogous bounds can be formed for the inclusive value functions of the worker-optimal stable match.

Using Proposition 2 (ii) and Lemma 4 we can then show that,

$$\begin{aligned}\bar{U}^M_{ijt}[\bar{U}, \bar{V}] &\geq \int \log\left(\exp\left\{\beta \bar{U}^M_{i0t+1}[\bar{U}, \bar{V}]\right\} + I^M_{wit+1}[\bar{U}, \bar{V}]\right) w(x_{it+1}|x_{it}, z_{jt}) dx_{it+1} + o_p(1) \\ &\geq \int \log\left(\exp\left\{\beta \bar{U}^M_{i0t+1}[\bar{U}, \bar{V}]\right\} + \hat{\Gamma}^M_{wt+1}[\bar{U}, \bar{V}](x_{it+1})\right) w(x_{it+1}|x_{it}, z_{jt}) dx_{it+1} + o_p(1).\end{aligned}$$

Similarly, we can show that

$$\bar{V}^M_{ijt}[\bar{U}, \bar{V}] \leq \int \log\left(\exp\left\{\beta \bar{V}^M_{0jt+1}[\bar{U}, \bar{V}]\right\} + \hat{\Gamma}^M_{mt+1}[\bar{U}, \bar{V}](z_{jt+1})\right) m(z_{jt+1}|x_{it}, z_{jt}) dz_{jt+1} + o_p(1),$$

where the sign the inequalities are reversed for a sequence of worker-optimal stable matches.

Combining these inequalities, we can show that for any vector of continuation values $(\bar{U}^*, \bar{V}^*)$ satisfying Assumption 3

$$\bar{U}^*_{ijt} \geq \bar{U}_t[\hat{\Gamma}^M_w[\bar{U}^*, \bar{V}^*]](x_{it}, z_{jt}) + o_p(1), \quad \textit{and} \quad \bar{V}^*_{ijt} \leq \bar{V}_t[\hat{\Gamma}^M_m[\bar{U}^*, \bar{V}^*]](x_{it}, z_{jt}) + o_p(1),$$

where the direction of the inequalities is reversed for the worker-optimal stable match. Overall, this implies that for any vector of continuation values $(\bar{U}^*, \bar{V}^*)$ satisfying Assumption 3:

$$\begin{aligned}&\hat{\Gamma}^M_{wt}(x) \geq \hat{\Psi}_{wt}[\hat{\Gamma}^M_w, \hat{\Gamma}^M_m](x) + o_p(1) \quad \text{and} \quad \hat{\Gamma}^M_{mt}(z) \leq \hat{\Psi}_{mt}[\hat{\Gamma}^M_w, \hat{\Gamma}^M_m](z) + o_p(1), \\ &\hat{\Gamma}^W_{wt}(x) \leq \hat{\Psi}_{wt}[\hat{\Gamma}^W_w, \hat{\Gamma}^W_m](x) + o_p(1) \quad \text{and} \quad \hat{\Gamma}^W_{mt}(z) \geq \hat{\Psi}_{mt}[\hat{\Gamma}^W_w, \hat{\Gamma}^W_m](z) + o_p(1),\end{aligned}$$

Given that $\hat{\Psi}_{wt}[\Gamma_w, \Gamma_m]$ and $\hat{\Psi}_{mt}[\Gamma_w, \Gamma_m]$ are nonincreasing in Γ_m and Γ_w, respectively, nondecreasing in Γ_w and Γ_m, respectively, and Lipschitz continuous in (Γ_w, Γ_m), we have:

$$\hat{\Gamma}^M_{wt} \geq \hat{\Psi}_{wt}[\hat{\Psi}_w[\hat{\Gamma}^M_w, \hat{\Gamma}^M_m], \hat{\Psi}_m[\hat{\Gamma}^M_w, \hat{\Gamma}^M_m]] + o_p(1)$$

Thus for any (Γ^*_w, Γ^*_m) solving the fixed point problem:

$$\Gamma^*_{wt} = \hat{\Psi}_{wt}[\Gamma^*_w, \Gamma^*_m] + o_p(1) \quad \text{and} \quad \Gamma^*_{mt} = \hat{\Psi}_{mt}[\Gamma^*_w, \Gamma^*_m] + o_p(1)$$

we thus have:

$$\hat{\Gamma}^M_{wt} \geq \Gamma^*_{wt} + o_p(1) \quad \text{and} \quad \hat{\Gamma}^M_{mt} \leq \Gamma^*_{mt} + o_p(1)$$

However, we know that the mapping $\hat{\Psi}$ is a contraction in logs, which means that it has a unique fixed point Γ^*. In addition, the firm-optimal stable match is unanimously preferred by firms while the worker-optimal stable match is unanimously preferred by workers (Roth and Sotomayor, 1992). This implies that for any $(\bar{U}, \bar{V})$, we have $M^M_{it}[\bar{U}, \bar{V}] \subset M^*_{it}[\bar{U}, \bar{V}] \subset M^W_{it}[\bar{U}, \bar{V}]$ and $W^W_{jt}[\bar{U}, \bar{V}] \subset W^*_{jt}[\bar{U}, \bar{V}] \subset W^M_{jt}[\bar{U}, \bar{V}]$ which means that for all i and j:

$$I^M_{wit}[\bar{U}, \bar{V}] \leq I^*_{wit}[\bar{U}, \bar{V}] \leq I^W_{wit}[\bar{U}, \bar{V}] \quad \text{and} \quad I^W_{mjt}[\bar{U}, \bar{V}] \leq I^*_{mjt}[\bar{U}, \bar{V}] \leq I^M_{mjt}[\bar{U}, \bar{V}].$$

This in turn implies that for all (x, z):

$$\hat{\Gamma}^M_{wt}(x) \leq \hat{\Gamma}^*_{wt}(x) \leq \hat{\Gamma}^W_{wt}(x) \quad \text{and} \quad \hat{\Gamma}^W_{mt}(z) \leq \hat{\Gamma}^*_{mt}(z) \leq \hat{\Gamma}^M_{mt}(z)$$

which implies that:

$$\Gamma^*_{wt} + o_p(1) \geq \hat{\Gamma}^W_{wt} \geq \hat{\Gamma}^M_{wt} \geq \Gamma^*_{wt} + o_p(1)$$

$$\Gamma^*_{mt} + o_p(1) \leq \hat{\Gamma}^W_{mt} \leq \hat{\Gamma}^M_{mt} \leq \Gamma^*_{mt} + o_p(1)$$

which in turn implies that:

$$\hat{\Gamma}^M_{wt} = \Gamma^*_{wt} + o_p(1) \quad \text{and} \quad \hat{\Gamma}^M_{mt} = \Gamma^*_{mt} + o_p(1)$$

$$\hat{\Gamma}^W_{wt} = \Gamma^*_{wt} + o_p(1) \quad \text{and} \quad \hat{\Gamma}^W_{mt} = \Gamma^*_{mt} + o_p(1)$$

Combining this with Lemma 4, this gives us for $(\bar{U}^*, \bar{V}^*)$ satisfying Assumption 3 and for all $i = 1, ..., n_w$ and all $j = 1, ..., n_m$:

$$I^M_{wit} = \Gamma^*_{wt} + o_p(1) \quad \text{and} \quad I^M_{mjt} = \Gamma^*_{mt} + o_p(1)$$

$$I^W_{wit} = \Gamma^*_{wt} + o_p(1) \quad \text{and} \quad I^W_{mjt} = \Gamma^*_{mt} + o_p(1)$$

which directly implies that for any $(\mu^*, \bar{U}^*, \bar{V}^*)$ satisfying Assumption 3:

$$I^*_{wit} = \Gamma^*_{wt} + o_p(1) \quad \text{and} \quad I^*_{mjt} = \Gamma^*_{mt} + o_p(1)$$

Lemma 5 finally implies uniform convergence of $\hat{\Gamma}$ to Γ^*. This completes the proof of convergence of inclusive values to Γ^*. Having established convergence of inclusive values, Proposition 1 and 2 directly imply Theorem 1.(ii) given the definition of F_n given in (1).

B Monte Carlo Simulations

To assess the validity of the asymptotic characterization described in Section 3 and the estimation strategy proposed in Section 4, I perform two Monte Carlo exercises. I simulate data from a market with an increasing number of participating agents to (i) verify whether empirical matching frequencies converge to their theoretical limit and (ii) verify that the ML estimator proposed is unbiased and consistent in large markets. I consider a market with three periods $T = 3$, normalize $\gamma_w = \gamma_m = 0$, and set $\beta = \delta = 0.9$. I specify the flow payoff functions as

$$U(x, z; \theta) = \theta_1 + \theta_2 z, \quad \textit{and} \quad V(x, z; \theta) = \theta_1 + \theta_3 x,$$

and set $\boldsymbol{\theta} = (1, 1, 1)$. I assume that $x_{i1} \sim \mathcal{N}(0, 1)$ and $z_{j1} \sim \mathcal{N}(0, 1)$ and that x_{it} and z_{jt} get incremented by one only if worker i and firm j are matched in period t, respectively. This simulates a setting where workers and firms become less attractive when they stay unmatched relative to forming a match.

To simulate the equilibrium, I first solve for inclusive values and continuation values by applying the sample analog of the contraction mappings described in (4) given the specified payoff functions, the initial aggregate distribution of states m_1 and w_1 and the transition distribution functions m and w. I then construct U_{ijt} and V_{ijt} given a simulated set of extreme value type-I shocks ϵ_{ijt} and η_{ijt}. Finally, I derive the worker-optimal stable match in each period t using the worker-proposing Deferred Acceptance algorithm.

In the first experiment, I simulate data from the DGP described above for different market sizes indexed by n and evaluate whether the observed matching frequencies converge to their theoretical limit. More specifically, I test whether the share of unmatched workers in each period converges to its limit. Table B.1 shows the results of this exercise. As the size of the market increases, the share of unmatched workers observed in the simulated data converges to its limit, which is displayed in the bottom row. However, the approximation error can be quite large (more than 2 percentage points) when the market size is relatively small ($n < 500$).

Table B.1: Monte Carlo: Share of Unmatched workers

n	$t=1$	$t=2$	$t=3$
20	0.2600	0.1870	0.2675
50	0.2439	0.1734	0.2447
100	0.2389	0.1640	0.2339
200	0.2314	0.1565	0.2237
500	0.2263	0.1509	0.2147
1000	0.2228	0.1469	0.2095
2000	0.2206	0.1432	0.2053
Model	0.2076	0.1384	0.1965

NOTES. This table reports the average share of unmatched firms and workers in each period taken over 200 sample draws for different sample sizes n.

Table B.2: Monte Carlo: MLE

n	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$
20	0.952 (0.475)	0.986 (0.352)	1.023 (0.334)
50	0.962 (0.292)	0.988 (0.223)	1.010 (0.204)
100	0.969 (0.192)	0.994 (0.156)	1.007 (0.140)
200	0.977 (0.133)	0.991 (0.104)	1.003 (0.105)
500	0.984 (0.088)	0.994 (0.067)	1.003 (0.063)
1000	0.992 (0.060)	0.995 (0.047)	1.002 (0.046)
θ	1	1	1

NOTES. This table reports the average and standard deviation of the ML estimator of $\boldsymbol{\theta}$ over 500 sample draws for different sample sizes n.

In the second experiment, I simulate data for different values of n and estimate $\boldsymbol{\theta}$ using the procedure described in Section 4. Table B.2 shows that the estimator features little bias even when the market size is small. When the market size increases, its bias and standard deviation vanish suggesting that it is consistent.

C Additional Tables and Figures

Table C.1: Participants to Centralized Recruitment Drive

	2015	2017	2019
# Applicants	64,070	71,717	61,687
Eligible Permanent	6,692	9,650	3,912
# Vacancies	36,257	40,942	33,322
Permanent	6,386	13,394	7,140
Temporary	29,871	27,548	26,182

NOTES. This Table displays the total number of applicants and vacancies participating in the national recruitment drive in 2015, 2017 and 2019.

Table C.2: Summary Statistics

	Mean	SD
Applicant Characteristics		
Test Score (std.)	0.060	1.011
Male	0.281	0.449
Experience > 5	0.342	0.474
Experience $= 0$	0.206	0.404
Job Characteristics		
Amenity Index	-0.274	1.172
Frontera	0.085	0.278
VRAEM	0.054	0.226
Multigrado	0.303	0.460
Unidocente	0.080	0.272
Bilingue	0.247	0.431
Travel Time	3.700	7.212
log(Pop)	8.039	3.801
Rural	0.621	0.485

NOTES. This Table shows the mean and standard deviation of the applicant and job characteristics used throughout the empirical analysis.

Table C.3: Estimates Teacher Value-Added Model

Panel A: Value-Added Coefficients	
μ_α	0.002 (0.007)
σ^2_α	0.147 (0.007)
Panel B: Student Characteristics	
Lagged Score	0.509 (0.003)
Lagged Score^2	-0.019 (0.002)
Female	-0.076 (0.004)
Quechua	-0.053 (0.015)
Age	-0.136 (0.005)
School Lagged Score	0.340 (0.047)
School Lagged Score^2	0.113 (0.044)
School Female	-0.352 (0.224)
School Quechua	0.640 (0.268)
School Age	-0.530 (0.117)
Class Lagged Score	-0.229 (0.027)
Class Lagged Score^2	-0.072 (0.023)
Class Female	0.446 (0.121)
Class Quechua	-0.515 (0.280)
Class Age	-0.026 (0.064)
Class Size	-0.003 (0.005)
Class Size^2	-0.000 (0.000)
Amenity Index	0.035 (0.020)
Frontera	0.103 (0.119)
Multigrado	-0.044 (0.069)
Unidocente	-0.298 (0.192)
Travel Time	-0.031 (0.084)
Travel Time^2	0.004 (0.004)
log(Pop)	-0.032 (0.041)
log(Pop)^2	0.001 (0.002)
Travel Time $\times$ log(Pop)	-0.004 (0.010)

NOTES. Panel A shows estimates of the distribution of the teacher value-added coefficients. Panel B displays the estimates of the parameters associated with student, school and classroom characteristics in the teacher value-added model. Standard errors are in parentheses.

Table C.4: Transition Process Test Scores

	Mean	Variance
	(1)	(2)
Intercept	22.775 (0.409)	22.571 (0.287)
Test Score	0.884 (0.003)	-0.059 (0.002)
Male	1.552 (0.165)	-0.272 (0.117)
Experience > 5	-1.077 (0.171)	0.293 (0.121)
Experience $= 0$	-0.683 (0.193)	-0.365 (0.137)
$\hat{\alpha}_i^*$	4.883 (0.884)	-0.428 (0.633)
Missing $\hat{\alpha}_i^*$	-3.763 (0.264)	-0.210 (0.187)
Urban School	1.633 (0.230)	-0.071 (0.161)

NOTES. This table displays estimates of the coefficients of the transition process described in equation (10) in Section 6.1.

Table C.5: Teachers Preference Estimates: Unobserved Heterogeneity

	× Group 2	× Group 3
	(1)	(2)
Intercept	1.183 (1.432)	-1.298 (1.209)
Wage	-2.590 (0.768)	-0.074 (0.522)
Amenity Index	0.010 (0.063)	-0.201 (0.079)
Distance Spline: 0-20km	-0.020 (0.009)	-0.033 (0.013)
Distance Spline: 20-100km	0.010 (0.002)	0.007 (0.003)
Distance Spline: 100km+	0.003 (0.000)	-0.003 (0.001)
Time to Travel	-0.667 (0.163)	0.109 (0.107)
Time to Travel^2	-0.008 (0.003)	-0.000 (0.001)
log(Pop)	-0.001 (0.157)	0.282 (0.190)
log(Pop)^2	0.038 (0.009)	-0.014 (0.012)
Time to Travel × log(Pop)	0.108 (0.019)	-0.013 (0.017)
Frontera	0.097 (0.255)	-0.508 (0.261)
VRAEM	0.067 (0.452)	-0.182 (0.383)
Multigrado	-0.107 (0.237)	-0.262 (0.216)
Unidocente	0.238 (0.364)	-0.113 (0.323)
Bilingue	0.342 (0.192)	0.265 (0.172)

NOTES. This table displays estimates of the parameters Θ_3 and Θ_4 described in equation (8). Standard errors are in parentheses.